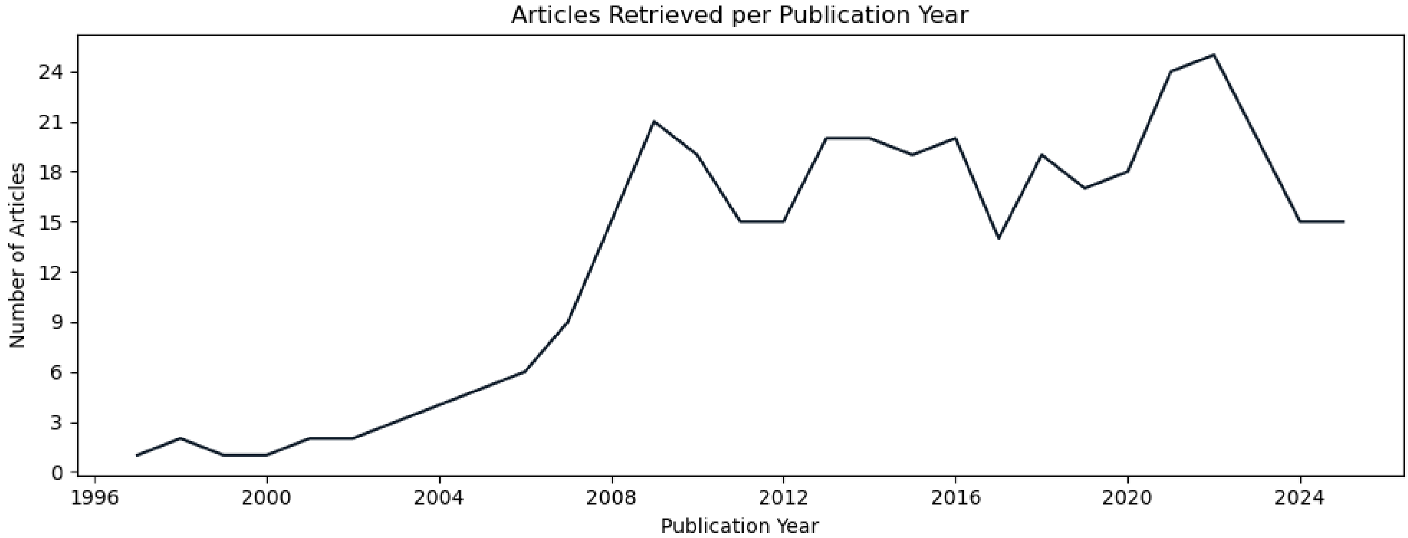
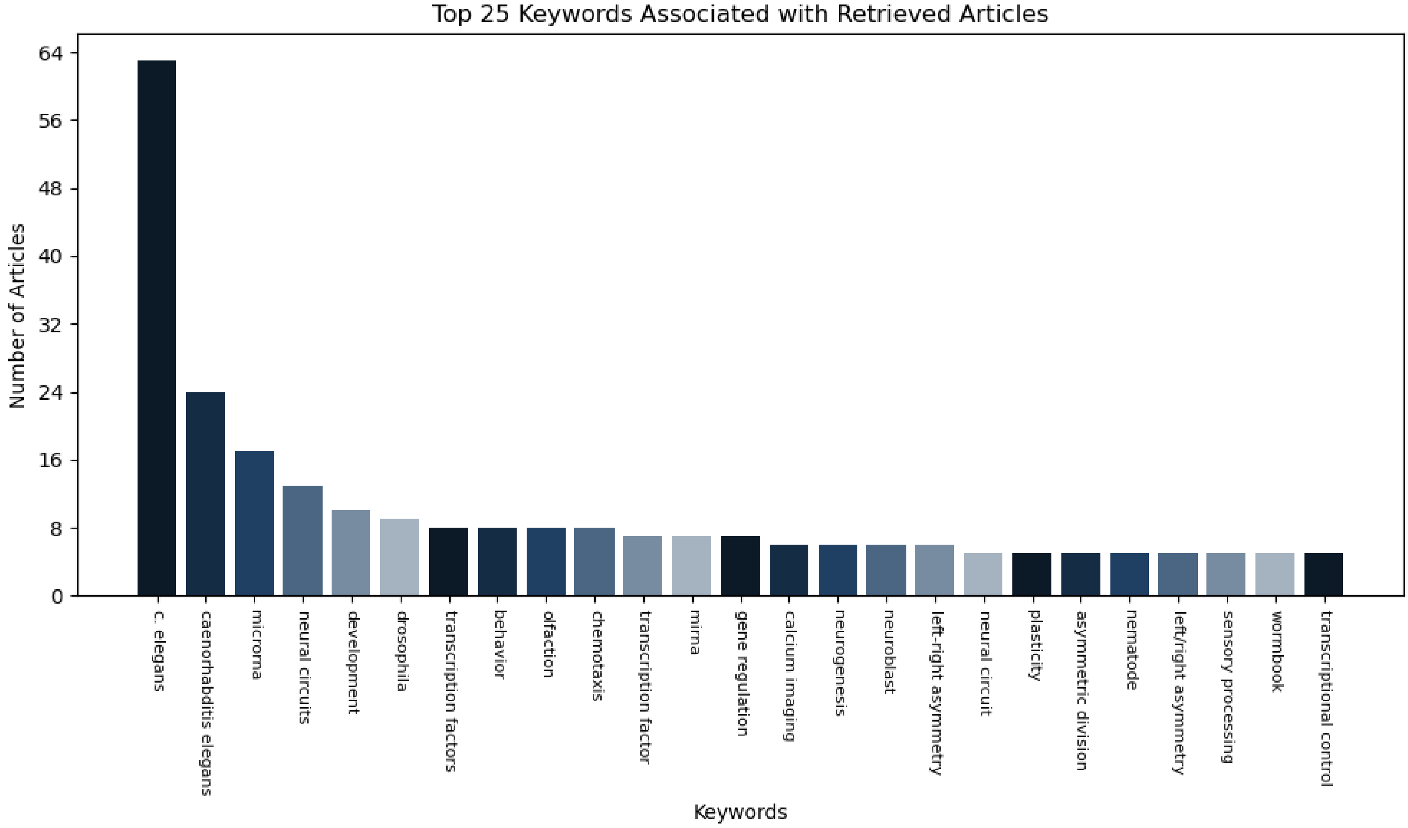
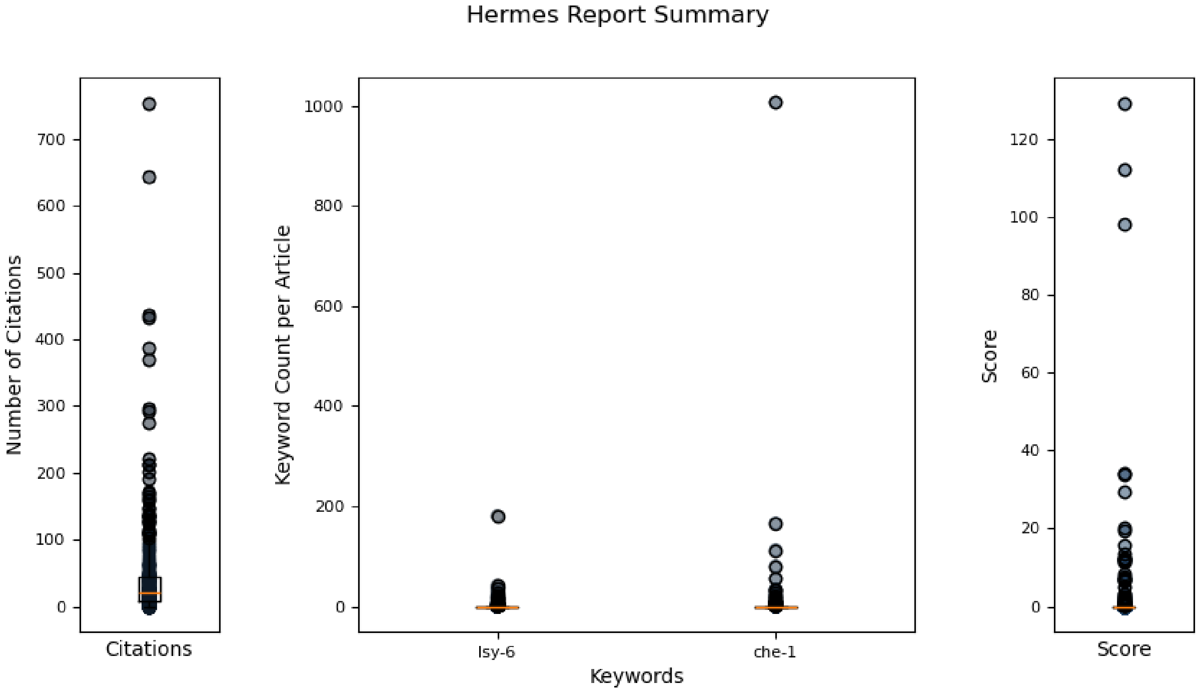
Literature Mining Report

Query Statistics

Query Date: 2025-12-18
Query Terms: "ASE asymmetry c elegans"
Scoring Keywords: che-1; lsy-6 (Default Mode)
Articles Parsed: 371/371 (100.00%)
Keyword Rarity Score (IDF): che-1: 2.179; lsy-6: 2.515







Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
34908528	2021	eLife	Traets, Joleen et al.	mechanism of life-long maintenance of neuron ...	9	129.56
33002421	2020	Developmental C	Charest, Julien et al.	combinatorial action of temporally segregated...	27	112.52
28422646	2017	eLife	Patel, Tulsi et al.	coordinated control of terminal differentiati...	47	98.37
18085825	2007	PLoS Genetics	Miska, Eric A et al.	most caenorhabditis elegans micrornas are ind...	297	34.31
40586702	2025	eLife	NA	evolution of lateralized gustation in nematod...	2	33.78
39282255	2025	bioRxiv	Mackie, Marisa et al.	evolution of lateralized gustation in nematod...	0	29.51
24065887	2013	Frontiers in Ce	Alqadah, Amel et al.	microrna function in left-right neuronal asym...	11	20.35
21698137	2011	PLoS Genetics	Poole, Richard et al.	a genome-wide rnai screen for factors involve...	33	19.57
31526477	2019	eLife	Hong, Ray L et al.	evolution of neuronal anatomy and circuitry i...	35	15.88
31584932	2019	PLoS Genetics	Li, Li et al.	hrpk-1, a conserved kh-domain protein, modula...	12	13.69
18560547	2008	PLoS ONE	Frkjr-Jensen, et al.	ammonium-acetate is sensed by gustatory and o...	18	12.57
19259264	2009	PLoS ONE	Tursun, Baris et al.	a toolkit and robust pipeline for the generat...	124	12.39
35504914	2022	Scientific Repo	Hebbar, Shilpa et al.	functional identification of microrna-centere...	2	12.16
25032706	2014	PLoS Genetics	Perales, Robert et al.	lin-42, the caenorhabditis elegans period hom...	29	12.02
27095199	2016	Nucleic Acids R	Flamand, Mathie et al.	poly(a)-binding proteins are required for mic...	23	11.59
19875417	2010	Nucleic Acids R	Takayama, Jun et al.	single-cell transcriptional analysis of taste...	18	11.43
26635721	2015	Frontiers in Ne	Davis, Gregory et al.	micrornas: not fine-tuners but key regulato...	42	8.26
29481550	2018	Nature methods	Alberti, Chiara et al.	cell-type specific sequencing of micrornas fr...	51	7.60
37027463	2023	Science Advance	Gleason, Ryan J et al.	developmentally programmed histone h3 express...	5	7.05
36329026	2022	Nature Communic	Chrystal, Paul et al.	the inner junction protein cfap20 functions i...	16	6.86
25569839	2015	Autophagy	Zhang, Hong et al.	guidelines for monitoring autophagy in caenor...	98	6.67
39167071	2024	Genetics	Poole, Richard et al.	Neurogenesis in Caenorhabditis elegans	11	6.30
28180320	2017	Nucleic Acids R	Wang, Chris et al.	teg-1 cd2bp2 controls mirna levels by regulat...	8	5.02
38421867	2024	Cell reports	Ray, Sneha et al.	neuron cilia restrain glial kcc-3 to a microd...	9	3.32
34504291	2021	Communications	Dekkers, Martij et al.	plasticity in gustatory and nociceptive neuro...	8	3.24

Query Result 1/25

Mechanism of life-long maintenance of neuron identity despite molecular fluctuations

Traets, Joleen JH; van der Burght, Servaas N; Rademakers, Suzanne; Jansen, Gert; van Zon, Jeroen S; Hauf, Silke; Walczak, Aleksandra M; Hauf, Silke; Becskei, Attila; Hauf, Silke

eLife

Year: 2021 PMID: 34908528 Citations: 9 Score: 129.56 DOI: 10.7554/eLife.66955.sa0

Keywords Hits: lsy-6: 0, che-1: 1006

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.044

Abstract:

Cell fate is maintained over long timescales, yet molecular fluctuations can lead to spontaneous loss of this differentiated state. Our simulations identified a possible mechanism that explains life-long maintenance of ASE neuron fate in Caenorhabditis elegans by the terminal selector transcription factor CHE-1. Here, fluctuations in CHE-1 level are buffered by the reservoir of CHE-1 bound at its target promoters, which ensures continued che-1 expression by preferentially binding the che-1 promoter. We provide experimental evidence for this mechanism by showing that che-1 expression was resilient to induced transient CHE-1 depletion, while both expression of CHE-1 targets and ASE function were lost. We identified a 130 bp che-1 promoter fragment responsible for this resilience, with deletion of a homeodomain binding site in this fragment causing stochastic loss of ASE identity long after its determination. Because network architectures that support this mechanism are highly conserved in cell differentiation, it may explain stable cell fate maintenance in many systems.

Summary:

Introduction: In most animal tissues, terminally differentiated cells are renewed on timescales of days to months yet the nervous system is unique, as most neurons hardly renew at all. Many mature neuron types exist that differ in the expression of hundreds of type-specific genes. How neuronal cells maintain this terminally different state over such long timescale decades, in the case of humans is not understood. Differentiation of neuron type is often controlled by a small subset of transcription factors, sometimes even a single one, that are called terminal selectors. These act through a conserved network motif called a Single-Input Module they bind to specific cis-regulatory control elements to induce both their own expression and that of the downstream target genes that define the neuronanal type. Such terminal selectr networks have been found to underlie differentiation of several neuron kinds in the nematode Caenorhabditis elegans ( ), photoreceptor subtypes in Drosophila ( ) and dopaminergic neurons in mice indicating that they form an evolutionary conserved principle for neuronion type determination.

Results: Loss of ASE neuron fate upon transient CHE-1 depletion.Auxin-inducible degradation system che-1 GFP AID.Anchorage larvae were exposed to 1 mM auxin to induce CHE-2 GFP depletions. CHE-3 GFP was strongly reduced after 1.5 hr, and undetectable after 3 h exposure to auxin.Aquatic exposure reduced chemotaxis to NaCl.

Discussion: The terminal selector gene che-1 controls neuron type determination of the salt-sensing ASE neurons, by inducing the expression of hundreds of ASE-specific target genes while also inducING its own expression via autoregulation. A previous study showed that upon inhibition of positive CHE-1 autoregulation, transient che-2 induction did not result in sustained che-3 expression, with ASE cell fate lost shortly after its determination. This result left open whether this positive feedback loop by itself is sufficient to maintain che-4 expression and ASE type for the animals entire lifetime, or whether additional mechanisms, such as chromatin modifications, are required after initial determination to lock in cell identity in an irreversible manner. Here, we show that transient depletion of CHE-2 is sufficient, indicating that indeed che-5 expression forms a reversible, bistable switch. This also raised the question how, in the presence of the inherent molecular fluctuations in the cell, this switch maintains its ON state (che-5) and prevents spontaneous transitions to the OFF state (noche-5 expression).

Figures:

Figure 1: Loss of ASE neuron fate upon transient CHE-1 depletion.

Figure 2: CHE-1 recovery after depletion.

Figure 3: CHE-1 depletion and NaCl chemotaxis.

Figure 4: che-1 mRNA and protein copy numbers in ASE neurons.

Figure 5: CHE-1 copy numbers in larvae and embryos.

Figure 6: che-1 mRNA and protein lifetimes.

Figure 7: Stable ON state by preferential binding of CHE-1 to its own promoter.

Figure 8: Dependence of ON state stability on number of targets and cooperativity.

Figure 9: Maintenance of che-1 expression during transient CHE-1 depletion.

Figure 10: Region flanking CHE-1-binding site ensures resilience to CHE-1 depletion.

Figure 11: Controls for promoter region mutants.

Figure 12: An Otx -related homeodomain transcription factor binding site involved in long-term maintenance of ASE cell fate.

Figure 13: Role of HD-TF-binding site in maintaining CHE-1::GFP expression.

Figure 14: Models of HD-TF action.

Mentioned biomedical entities:

Genes: 2B, ADF, ALR-1, ASE, ASER, ASK, B, CEH-36, CHE-1, Che-1, EGL-43, GAL, Gal4, NHR-67, OSM-3, P40, TF, ceh-23, cog-1, dsc-1, g51, g55, g57, gcy-22, histone, lim-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: E. coli, humans, mice

Tissues: neuron, tissue

Pathways: N/A

Query Result 2/25

Combinatorial Action of Temporally Segregated Transcription Factors

Charest, Julien; Daniele, Thomas; Wang, Jingkui; Bykov, Aleksandr; Mandlbauer, Ariane; Asparuhova, Mila; Rhsner, Josef; Gutirrez-Prez, Paula; Cochella, Luisa

Developmental Cell

Year: 2020 PMID: 33002421 Citations: 27 Score: 112.52 DOI: 10.1016/j.devcel.2020.09.002

Keywords Hits: lsy-6: 180, che-1: 55

Keywords Frequency (TF): lsy-6: 0.014; che-1: 0.004

Abstract:

Combinatorial action of transcription factors (TFs) with partially overlapping expression is a widespread strategy to generate novel gene-expression patterns and, thus, cellular diversity. Known mechanisms underlying combinatorial activity require co-expression of TFs within the same cell. Here, we describe the mechanism by which two TFs that are never co-expressed generate a new, intersectional expression pattern in C.elegans embryos: lineage-specific priming of a gene by a transiently expressed TF generates a unique intersection with a second TF acting on the same gene four cell divisions later; the second TF is expressed in multiple cells but only activates transcription in those where priming occurred. Early induction of active transcription is necessary and sufficient to establish a competent state, maintained by broadly expressed regulators in the absence of the initial trigger. We uncover additional cells diversified through this mechanism. Our findings define a mechanism for combinatorial TF activity with important implications for generation of cell-type diversity.

Summary:

Introduction: TFs are a key component of the development of neurons. Combinatorial use of TF enables the generation of novel, specific gene-expression patterns that exceed the number of available TF. The specification of individual cell identities by combinations of CFs is a basis for cellular diversification during development. The concept of temporal intersection in which two TF are required for transcriptional activation, but separated in time. TF1 is expressed transiently in a subset of progenitor cells, while TF2 is expressed in a specialized subset that is undergoing differentiation.

Results: Direct, early binding of TBX-37/38 is necessary and sufficient to establish competence for transcription in the post-mitotic ASEL Neurons. To begin to dissect the mechanism by which TBx-37 and tbx-38 contribute to the specific expression oflsy-6in the post mitotic ASE, we explored the possibility that these TFs bind directly to thelsi-6locus already at the 28-cell stage, when they are first detected.

Discussion: We show that two temporally segregated TFs act combinatorially to achieve cell-type specific transcription of a locus that controls neuronal identity. We also provide evidence that temporal integration of TF may be a more general mechanism for cellular diversification. The use of TBX-37/38 for gene activation is a widespread phenomenon that has been long recognized to enable generation of cell- type-specific gene-expression patterns and, in turn, cell-typal diversity. Here, we revealed mechanistic insight into how two temporally segregate TF combinations can act combinatively to achieve cellular specific transcription.

Figures:

Figure 1: Integration of Transcriptional Inputs over Time as a Mechanism for Cell Diversification

Figure 2: Early, Direct Binding of TBX-37/38 to lsy-6 Is Necessary and Sufficient to Establish Competence for Transcription in the ASE Neurons

Figure 3: TBX-37/38 Are Not Continuously Required for lsy-6 Expression

Figure 4: TBX-37/38 Establish a Differentially Accessible State of the lsy-6 Locus

Figure 5: Transcription over the lsy-6 Locus Occurs Bidirectionally and Is Required for Priming

Figure 6: Asymmetric lsy-6 Competence Is Maintained in a Symmetric Trans -acting Factor Environment

Figure 7: TBX-37/38 Determine Left-Right Molecular Asymmetry across Multiple Bilateral Neuron Pairs with ABa/ABp Lineage Origin

Mentioned biomedical entities:

Genes: ASE, ASER, AVE, CHE-1, Cas9, FoxA, Mos1, P40, PHA-4, SEP, SM2, TBX-37, TBX-38, TEL, TF, TF1, VP16, ZIF-1, Zdbf2, cat, galK, gcy-5, histone, lsy-6, unc-119

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: E. coli, E.coli, cat, mouse

Tissues: capillary, egg, eggshell, neuron, tube

Pathways: N/A

Query Result 3/25

Coordinated control of terminal differentiation and restriction of cellular plasticity

Patel, Tulsi; Hobert, Oliver; Shen, Kang; Shen, Kang

eLife

Year: 2017 PMID: 28422646 Citations: 47 Score: 98.37 DOI: 10.7554/eLife.24100.015

Keywords Hits: lsy-6: 0, che-1: 166

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.014

Abstract:

The acquisition of a specific cellular identity is usually paralleled by a restriction of cellular plasticity. Whether and how these two processes are coordinated is poorly understood. Transcription factors called terminal selectors activate identity-specific effector genes during neuronal differentiation to define the structural and functional properties of a neuron. To study restriction of plasticity, we ectopically expressed C. elegans CHE-1, a terminal selector of ASE sensory neuron identity. In undifferentiated cells, ectopic expression of CHE-1 results in activation of ASE neuron type-specific effector genes. Once cells differentiate, their plasticity is restricted and ectopic expression of CHE-1 no longer results in activation of ASE effector genes. In striking contrast, removal of the respective terminal selectors of other sensory, inter-, or motor neuron types now enables ectopically expressed CHE-1 to activate its ASE-specific effector genes, indicating that terminal selectors not only activate effector gene batteries but also control the restriction of cellular plasticity. Terminal selectors mediate this restriction at least partially by organizing chromatin. The chromatin structure of a CHE-1 target locus is less compact in neurons that lack their resident terminal selector and genetic epistasis studies with H3K9 methyltransferases suggest that this chromatin modification acts downstream of a terminal selector to restrict plasticity. Taken together, terminal selectors activate identity-specific genes and make non-identity-defining genes less accessible, thereby serving as a checkpoint to coordinate identity specification with restriction of cellular plasticity.

Summary:

Introduction: The acquisition of differentiated cell identities in a multicellular organism is accompanied by the loss of developmental plasticity. Undifferentiated cells are pluripotent and have the potential to activate diverse categories of genes. As differentiation progresses, cells activate cellular identity-specific genes as well as repress non-identity-specific Genes, thereby losing pluriportency and attaining a differentiated state that is restricted. In this work, we aimed to uncover molecules and mechanisms that coordinate identity specification with the restriction of cellular plasticity

Results: Differential ability of CHE-1 to activate target genes in C.elegans. To gain insights into the mechanisms that restrict plasticity, we probed cellular plasticity in many cell types of C.elggansthroughout different developmental stages. To this end, we ubiquitously expressed Zn-finger TF CHE--1 at different time points in developing worms. CHE is normally expressed only in the ASEs, two sensory neurons located in the head, and acts as a terminal selector by activating expression of numerous ASE-specific effector genes).

Discussion: Differentiated cells are defined by identity-specific transcriptional programs that are stably maintained throughout the life of a cell. Here, we provide evidence that in C. elegans neurons, identity-defining terminal selectors not only initiate and maintain the expression of effector genes that define a cellular phenotype but they also trigger the restriction of cellular plasticity. We propose that the coordination of identity specification and restriction of plasticity ensures the robustness of the differentiated state, protecting cells from accidental transcription of unwanted genes in response to fluctuations in the environment.

Figures:

Figure 1: Expression of ASE markers in response to CHE-1 hs at different developmental stages.

Figure 2: Tools used for the detection of gcy-5 expression.

Figure 3: Expression of gcy-5 in response to CHE-1 hs at different developmental stages.

Figure 4: Expression of ASE markers in response to cell type-specific, but not temporally controlled, che-1 expression.

Figure 5: Several CHE-1 targets retaintheirCHE-1responsiveness in unc-3 mutant cholinergic MNs.

Figure 6: Several distinct terminal selectors control restriction of cellular plasticity in distinct neuron types.

Figure 7: Phenotypes of additional mutants.

Figure 8: Geneticseparation of identity-specification and plasticityrestriction.

Figure 9: Assessing chromatin state of a gcy-5 transgenic locus withthe LacI/LacO dot assay.

Figure 10: Interactions of unc-3 with chromatin mutants.

Figure 11: H3K9 in wildtype and unc-3 mutant worms.

Figure 12: A model for the role of terminal selectors in restriction of plasticity.

Mentioned biomedical entities:

Genes: ASE, ASER, ASK, B, C, CHE-1, COE, D, HA, HP1, HPL-1, HPL-2, Klf4, MBL, MET-2, Oct4, PAG, PRC2, Pax5, RIG, Sox2, TF, TIR1, TP, UNC-3, VNC, elt-3, elt-7, gcy-5, gcy-6, histone, hlh-1, hnd-1, hpl-1, hsp16, lim-6, lin-11, ttx-1, unc-120, unc-3, unc-30, unc-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, fibroblast, neuron

Organisms: C. elegans, C.elegans, E. coli, humans, mice

Tissues: cuticle, hypodermis, neuron

Pathways: N/A

Query Result 4/25

Most Caenorhabditis elegans microRNAs Are Individually Not Essential for Development or Viability

Miska, Eric A; Alvarez-Saavedra, Ezequiel; Abbott, Allison L; Lau, Nelson C; Hellman, Andrew B; McGonagle, Shannon M; Bartel, David P; Ambros, Victor R; Horvitz, H. Robert; McManus, Michael T

PLoS Genetics

Year: 2007 PMID: 18085825 Citations: 297 Score: 34.31 DOI: 10.1371/journal.pgen.0030215

Keywords Hits: lsy-6: 4, che-1: 0

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.000

Abstract:
MicroRNAs (miRNAs), a large class of short noncoding RNAs found in many plants and animals, often act to post-transcriptionally inhibit gene expression. We report the generation of deletion mutations in 87 miRNA genes in Caenorhabditis elegans, expanding the number of mutated miRNA genes to 95, or 83% of known C. elegans miRNAs. We find that the majority of miRNAs are not essential for the viability or development of C. elegans, and mutations in most miRNA genes do not result in grossly abnormal phenotypes. These observations are consistent with the hypothesis that there is significant functional redundancy among miRNAs or among gene pathways regulated by miRNAs. This study represents the first comprehensive genetic analysis of miRNA function in any organism and provides a unique, permanent resource for the systematic study of miRNAs.

Summary:
Introduction: MicroRNAs (miRNAs) were discovered in C. elegans during studies of the control of developmental timing. miRNAs are approximately 22-nucleotide noncoding RNAs that are thought to regulate gene expression through sequence-specific base-pairing with target mRNAs. miRNs have been identified in organisms as diverse as roundworms, flies, fish, pygmy frogs, mammals, flowering plants, mosses, and even viruses, using genetics, molecular cloning, and predictions from bioinformatics. In C.eleganson about 115 miRNA genes have been confidently identified ,.

Results: The cloning of many miRNAs fromC. elegans using molecular biological techniques prompted us to take a genetic approach to study miRNA function in vivo inC.eleganc. egansthrough the generation of loss-of-function mutants. We isolated deletion mutants using establishedC.egancc.eiganstechniques. The public database for miRNA, miRBase release 9.0, listed 114C.

Discussion: Here we report the first large-scale collection of miRNA loss-of-function mutants for any organism. We isolated new deletion alleles for 87 miRNA genes. Together with two publicly available deletion mutants, three mutants that we described elsewhere, and three mutant generated in genetic screens, there are now mutants . We hope that this collection will become a widely used resource for the study of miRN function. We also report the overexpression of miR-84 and miR-60 from transgenes inC. elegansaffects vulval development ,. The overexpression , of mir-84, suppresses the multivulva phenotype oflet-60 RASactivation mutants. Iflet-60

Figures:

Mentioned biomedical entities:
Genes: ADL, AGO, AL, ASK, Argonaute, Dicer, NCL, cog-1, hbl-1, hid, let-7, lin-14, lin-28, lin-4, lsy-6, miR-14, miR-84, mir-7, mir-84

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: 87, C. elegans, fish, frogs, mice, mouse, zebrafish

Tissues: Egg, seed, tissue

Pathways: N/A

Query Result 5/25

Evolution of lateralized gustation in nematodes

NA

eLife

Year: 2025 PMID: 40586702 Citations: 2 Score: 33.78 DOI: 10.7554/eLife.103796.3.sa0

Keywords Hits: lsy-6: 7, che-1: 112

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.012

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in Pristionchus pacificus respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode Caenorhabditis elegans. Previously we showed that P. pacificus likely lacked bilateral asymmetry (Hong et al., 2019). Here, we show that by visualizing neuronal activity patterns, contrary to previous expectations based on its genome sequence, the salt responses of P. pacificus are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Introduction: Nematodes form a vast array of ecological relationships, from specialized parasitehost dependencies to nematodemicrobial interactions, each one demanding exquisitely fine-tuned sets of sensory palates that span multiple modalities. Yet, the number of sensory neurons across diverse nemode species seems to be constrained. To identify the genetic basis for patterning different nervous systems and to understand the processes that underlie evolutionary changes in adapting to different environments, several comparative model systems have been developed to promote comparisons to the well-studied nematodeCaenorhabditis elegansat the genetic and cellular levels, including the predatory entomophilic nemate,Pristionchus pacificus.

Results: Comparing the chemosensory profiles of P. pacificus and C. elegans suggests differences in their salt preferences. NH4I is more attractive to P.pacificu than C.elegan. Similarly, P. Pacificus is less attracted to NaCl and LiCl than to ammonium salts. Interestingly, P pacifius is repulsed by acetate salts, which induce attractive responses in C.elgans.

Discussion: Our study has revealed several insights into the substrates of evolutionary changes between two distantly related nematode species, P. pacificus and C. elegans. We used intracellular calcium levels as our readout for neuronal activity with a genetically encoded calcium sensor in two pairs of che-1-expressing amphid sensory neurons the first calcium imaging study in P. Pacificus. We show that three neuron types (ASE left, ASE right, and AFD neurons) each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. We have identified the first laterally asymmetric marker in P . Pacificus,gcy-22.3p GFP, with its expression limited to ASER homolog (AM7). Our unexpected discovery that neuron asymmetry is present in the ASE homologs between two closely related nematodes, despite the lack of a alsy-6 homolog, suggests that functional lateralization in P

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions between P. pacificus and C. elegans .

Figure 2: P. pacificus che-1 expression in ASE and AFD amphid neurons.

Figure 3: Co-expression of gcy-22.3p::GFP and ttx-1p::RFP in ASER.

Figure 4: che-1 expressing amphid neurons are required for sensing water-soluble ions in P. pacificus .

Figure 5: The che-1p::HisCl1 transgene is necessary for histamine-dependent knockdown of salt attraction.

Figure 6: Molecular lesions for P. pacificus che-1 and gcy 22.3 .

Figure 7: ASEL and ASER responses to different concentrations of NH 4 Cl, NaCl, and NH 4 I.

Figure 8: Negative controls of ASE and AFD neuron responses to green light.

Figure 9: Individual traces and heatmaps of ASE responses to various salts.

Figure 10: Microfluidic apparatus used for calcium imaging.

Figure 11: Combined AFD responses in comparison to ASE left or right neuron responses to NH 4 Cl, NaCl, and NH 4 I.

Figure 12: Individual traces and heatmaps of AFD responses to various salts.

Figure 13: AFDL/R responses to different concentrations of NH 4 Cl, NaCl, and NH 4 I.

Figure 14: The laterally asymmetric expression of gcy-22.3 is dependent on the zinc finger transcription factor CHE-1.

Figure 15: ASE and AFD responses in wildtype compared to gcy-22.3 mutants.

Figure 16: AFDL and AFDR responses to NH 4 Cl and NaCl in wildtype and gcy-22.3 mutant.

Figure 17: Chemotaxis responses to individual ions in gcy-22.3 mutants.

Mentioned biomedical entities:

Genes: ADF, ALFA, ASE, ASER, ASK, B, CHE-1, Cas9, Cel, HisCl1, IMD, MA, Max, Notch, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy, gcy-22, gcy-8, lim-6, lin-4, lsy-6, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, human

Tissues: neuron, ventral

Pathways: N/A

Query Result 6/25

Evolution of lateralized gustation in nematodes

Mackie, Marisa; Le, Vivian Vy; Carstensen, Heather R.; Kushnir, Nicole R.; Castro, Dylan L.; Dimov, Ivan M.; Quach, Kathleen T.; Cook, Steven J.; Hobert, Oliver; Chalasani, Sreekanth H.; Hong, Ray L.

bioRxiv

Year: 2025 PMID: 39282255 Citations: 0 Score: 29.51 DOI: 10.1101/2024.08.31.610597

Keywords Hits: lsy-6: 5, che-1: 80

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.013

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in Pristionchus pacificus respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode Caenorhabditis elegans. By visualizing neuronal activity patterns, we show that contrary to previous expectations based on its genome sequence, the salt responses of P. pacificus are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Results: P. pacificus and C. elegans display differences in their salt preferences. Previous cross-species comparisons betweenP. eliotus andC. ergansindicated strong differences in olfaction preferences that reflect their divergent evolutionary histories and ecology. To identify the neurons that mediate gustation, we first compared the chemosensory profiles of these two species toward water-soluble ions. In this survey, we found ammonium salts to be the strongest attractants to wildtypeP.

Discussion: Our study has revealed several insights into the substrates of evolutionary changes between two distantly related nematode species, P. pacificus and C. elegans. We used intracellular calcium levels as our readout for neuronal activity with a genetically encoded calcium sensor in two pairs of che-1-expressing amphid sensory neurons the first calcium imaging study in P. Pacificus. We show that three neuron types (ASE left, ASE right and AFD neurons) each have distinct calcium responses to specific ion concentrations, revealing diversity at the single neuron level. We have identified the first laterally asymmetric marker in P . Pacificus,gcy-22.3p GFP,with its expression limited to ASER homolog (AM7). Our unexpected discovery that neuron asymmetry is present in the ASE homologs between two closely related nematodes, despite the lack of a alsy-6 homolog, suggests that functional lateralization in P

Figures:

Figure 1: A comparison of chemotaxis responses to water-soluble ions between P. pacificus and C. elegans .

Figure 2: P. pacificus che-1 expression in ASE and AFD amphid neurons.

Figure 3: che-1 expressing amphid neurons are required for sensing water-soluble ions in P. pacificus .

Figure 4: ASEL and ASER responses to different concentrations of NH 4 Cl, NaCl, and NH 4 I.

Figure 5: Combined AFD responses in comparison to ASE left or right neuron responses to NH 4 Cl, NaCl, and NH 4 I.

Figure 6: The laterally asymmetric expression of gcy-22.3 is dependent on the zinc finger transcription factor CHE-1.

Figure 7: ASE and AFD responses in wildtype compared to gcy-22.3 mutants.

Figure 8: Chemotaxis responses to individual ions in gcy-22.3 mutants.

Mentioned biomedical entities:

Genes: ADF, ALFA, ASE, ASER, ASK, CA, CHE-1, Cas9, Cel, IMD, MA, Notch, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, fozi-1, gcy, lim-6, lin-4, lsy-6, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: C. elegans, human

Tissues: neuron, ventral

Pathways: N/A

Query Result 7/25

microRNA function in left-right neuronal asymmetry: perspectives from C. elegans

Alqadah, Amel; Hsieh, Yi-Wen; Chuang, Chiou-Fen

Frontiers in Cellular Neuroscience

Year: 2013 PMID: 24065887 Citations: 11 Score: 20.35 DOI: 10.3389/fncel.2013.00158

Keywords Hits: lsy-6: 20, che-1: 3

Keywords Frequency (TF): lsy-6: 0.008; che-1: 0.001

Abstract:

Leftright asymmetry in anatomical structures and functions of the nervous system is present throughout the animal kingdom. For example, language centers are localized in the left side of the human brain, while spatial recognition functions are found in the right hemisphere in the majority of the population. Disruption of asymmetry in the nervous system is correlated with neurological disorders. Although anatomical and functional asymmetries are observed in mammalian nervous systems, it has been a challenge to identify the molecular basis of these asymmetries. C. elegans has emerged as a prime model organism to investigate molecular asymmetries in the nervous system, as it has been shown to display functional asymmetries clearly correlated to asymmetric distribution and regulation of biologically relevant molecules. Small non-coding RNAs have been recently implicated in various aspects of neural development. Here, we review cases in which microRNAs are crucial for establishing leftright asymmetries in the C. elegans nervous system. These studies may provide insight into how molecular and functional asymmetries are established in the human brain.

Summary:

INTRODUCTION: MicroRNAs (miRNAs) are endogenous 2024 nt small non-coding RNAs that regulate gene expression through binding to complementary sequences in target messenger RNA (mRNAs), leading to translational repression and/or cleavage of target mRNA(s). While most miRNAs downregulate gene expression, there are examples of miRNA-mediated upregulation of target gene expression during cell cycle arrest, suggesting that miRNA function is complex and context dependent. miRNA have been implicated in many aspects of development and disease including cell cycle, cell differentiation, apoptosis, life span, developmental timing, stress responses, neural development and regeneration, cancers, and neurodegenerative disorders.

PERSPECTIVES: The role of miRNAs in establishing left-right asymmetry in the C. elegans nervous system can lay the groundwork for identifying the processes used in higher organisms, as the methods used may be evolutionarily conserved. Because of the highly conserved nature of miRBs, insights into how they are involved to control asymmetric fates will help facilitate our understanding in vertebrate neuronal asynchrony. The involvement of miRns in the asynchronous process may also be reflective of the principles these small non-coding RNAs use in directing other neurodevelopmental processes.

Figures:

Figure 1: Priming and boosting model of ASE asymmetry. ASEL: (1) The first step of promoting the ASEL cell fate occurs six cell divisions before the postmitotic ASEL is born. A transient pair of T-box transcription factors TBX-37/38 acts on the downstream primer element of lsy-6 miRNA to open up the chromatin. (2) The priming event allows boosting of lsy-6 miRNA levels by the CHE-1 transcription factor in the ASEL mother cell, leading to high levels of the miRNA in the postmitotic ASEL neuron. (3) The lsy-6 miRNA directly inhibits the ASER promoting transcription factor COG-1, and allows the ASEL promoting transcription factor DIE-1 to be expressed. (4) DIE-1 then activates ASEL effector genes and represses the ASER cell fate. ASER: (1) Notch is expressed in the ASER precursor and inhibits the lsy-6 priming event. Therefore, the lsy-6 miRNA expression level cannot be boosted as the chromatin remains compact. (2) Consequently, in the postmitotic ASER neuron, the COG-1 transcription factor is expressed, which activates another miRNA, mir-273 . (3) mir-273 directly inhibits the ASEL promoting transcription factor DIE-1, and therefore promotes the ASER fate.

Figure 2: Model of mir-71 function in AWC asymmetry. In the AWC OFF cell, high calcium level activates the calcium-regulated UNC-43/TIR-1/MAPK cascade, leading to srsx-3 expression. In the AWC ON cell, nsy-4 and nsy-5 promote the stability of mature mir-71 through an unknown mechanism. The mature mir-71 miRNA targets the 3UTR of tir-1 mRNA for degradation, leading to the inhibition of the calcium signaling pathway and the subsequent induction of str-2 expression. Orange color represents factors that promote AWC ON . Blue color represents factors that promote AWC OFF . Gray/white color represents inactive or less active factors. Dotted line represents factors being downregulated. PD, pentanedione; BU, butanone.

Mentioned biomedical entities:

Genes: ASE, ASER, CHE-1, COG-1, DIE-1, LIN-14, LMO4, Myt1, Notch, TRIM32, dicer, die-1, let-7, lin-4, lsy-6, miR-9a, mir-273, nsy-4, srsx-3, str-2, xrn-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron, progenitor cell

Organisms: human, humans, mice, zebrafish

Tissues: N/A

Pathways: N/A

Query Result 8/25

A Genome-Wide RNAi Screen for Factors Involved in Neuronal Specification in Caenorhabditis elegans, bScreen for Neuronal Fate Mutants

Poole, Richard J.; Bashllari, Enkelejda; Cochella, Luisa; Flowers, Eileen B.; Hobert, Oliver; Mango, Susan E.

PLoS Genetics

Year: 2011 PMID: 21698137 Citations: 33 Score: 19.57 DOI: 10.1371/journal.pgen.1002109

Keywords Hits: lsy-6: 35, che-1: 12

Keywords Frequency (TF): lsy-6: 0.003; che-1: 0.001

Abstract:
One of the central goals of developmental neurobiology is to describe and understand the multi-tiered molecular events that control the progression of a fertilized egg to a terminally differentiated neuron. In the nematode Caenorhabditis elegans, the progression from egg to terminally differentiated neuron has been visually traced by lineage analysis. For example, the two gustatory neurons ASEL and ASER, a bilaterally symmetric neuron pair that is functionally lateralized, are generated from a fertilized egg through an invariant sequence of 11 cellular cleavages that occur stereotypically along specific cleavage planes. Molecular events that occur along this developmental pathway are only superficially understood. We take here an unbiased, genome-wide approach to identify genes that may act at any stage to ensure the correct differentiation of ASEL. Screening a genome-wide RNAi library that knocks-down 18,179 genes (94% of the genome), we identified 245 genes that affect the development of the ASEL neuron, such that the neuron is either not generated, its fate is converted to that of another cell, or cells from other lineage branches now adopt ASEL fate. We analyze in detail two factors that we identify from this screen: (1) the proneural gene hlh-14, which we find to be bilaterally expressed in the ASEL/R lineages despite their asymmetric lineage origins and which we find is required to generate neurons from several lineage branches including the ASE neurons, and (2) the COMPASS histone methyltransferase complex, which we find to be a critical embryonic inducer of ASEL/R asymmetry, acting upstream of the previously identified miRNA lsy-6. Our study represents the first comprehensive, genome-wide analysis of a single neuronal cell fate decision. The results of this analysis provide a starting point for future studies that will eventually lead to a more complete understanding of how individual neuronal cell types are generated from a single-cell embryo.

Summary:

Introduction: RNAi technology in C.eleganshas opened an alternative path to classic forward analysis to study the genetic bases of various processes. However, this approach has not been undertaken in Celegansta to comprehensively study neuronal development and fate determination. We are using the ASE gustatory neuron class, composed of two bilaterally symmetric neurons, ASEL and ASER, as a paradigm to understand how neuroneal fate is specified. Each neuron is produced by two distinct descendants of the embryonic blastomere AB called ABa (generates ASEL) and ABp (generates ASER). There is a progressive restriction in the types of cells produced by the AB blastomerea. The great-granddaughter of AB, ABalp, still produces ASEL.

Results: RNAi is known to work less efficiently in the nervous system than in other tissues. Indeed, we find that dsRNA directed againstgfpbarely reducesgffexpression levels in transgenic animals that drive agfpreporter in the ASEL neuron Several genetic backgrounds have been described in which animals are more sensitive to RNAisensitized. We find that in all of them ASE-specificgfpopulation is now significantly reduced, withnre-1 lin-15banderi-1; li-15bmutants allowing more effectivegfknockdown thanrrf-3 10.1371/journal.pgen.1002109.g002Figure 2Testing hypersensitive strains and validating the screen with visible phenotypes.

Discussion: We describe the first comprehensive, genome-wide RNAi screen that monitors a single neuron-type cell fate decision. The screen has been remarkably successful in retrieving genes known to be involved in controlling ASE neuron specification. Six genes previously known tobe involved in ASE Neuron Specification have been found in an unbiased manner (che-1, die-1, lsy-2, lim-6, fozi-1 and only two have not (cog-1,nhr-67: these genes also display no laterality defects with sequence-verified RNAin clones). The RNAis screen also revealed a number of genes that we would have predicted to result in ASA lineage defects, including genes involved in the segregation of early patterning cues (e.g.parandmex-genes) and involved in binary fate decision system (lit-1gene).

Figures:

Figure 1: The embryonic origins and specification of the ASE neurons.

Figure 2: Testing hypersensitive strains and validating the screen with visible phenotypes.

Figure 3: Overview of the screening strategy.

Figure 4: Examples of phenotypic classes uncovered by the screen.

Figure 5: The achaete-scute homolog hlh-14 is required to specify neurons in the ASE lineage.

Figure 6: Asymmetric expression and function of hlh-14 in the C-lineage.

Figure 7: The evolutionary conserved methyltransferase COMPASS complex is required for ASE asymmetric cell fate specification.

Figure 8: The cloning and expression of lsy-15/rbbp-5(ot86) .

Mentioned biomedical entities:

Genes: ASE, ASER, Brd, Gro, LIN-49, Lon, MYST, NBP, Notch, P0, P3, PHD, RbBP5, Unc, WD40, WDR5, WDR82, ash-2, cog-1, die-1, dpy-7, fozi-1, histone, lim-6, lin-32, lsy-6, mex-5, nhr-67, pie-1, ref-1, sys-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, ectoderm, muscle cell, neuron

Organisms: C. elegans, C.elegans, human

Tissues: blastomere, cholinergic neuron, cuticle, egg, neuron, tissue, ventral

Pathways: N/A

Query Result 9/25

Evolution of neuronal anatomy and circuitry in two highly divergent nematode species

Hong, Ray L; Riebesell, Metta; Bumbarger, Daniel J; Cook, Steven J; Carstensen, Heather R; Sarpolaki, Tahmineh; Cochella, Luisa; Castrejon, Jessica; Moreno, Eduardo; Sieriebriennikov, Bogdan; Hobert, Oliver; Sommer, Ralf J; Sengupta, Piali; Marder, Eve; Sengupta, Piali

eLife

Year: 2019 PMID: 31526477 Citations: 35 Score: 15.88 DOI: 10.7554/eLife.47155.053

Keywords Hits: lsy-6: 4, che-1: 33

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.002

Abstract:

The nematodes C. elegans and P. pacificus populate diverse habitats and display distinct patterns of behavior. To understand how their nervous systems have diverged, we undertook a detailed examination of the neuroanatomy of the chemosensory system of P. pacificus. Using independent features such as cell body position, axon projections and lipophilic dye uptake, we have assigned homologies between the amphid neurons, their first-layer interneurons, and several internal receptor neurons of P. pacificus and C. elegans. We found that neuronal number and soma position are highly conserved. However, the morphological elaborations of several amphid cilia are different between them, most notably in the absence of winged cilia morphology in P. pacificus. We established a synaptic wiring diagram of amphid sensory neurons and amphid interneurons in P. pacificus and found striking patterns of conservation and divergence in connectivity relative to C. elegans, but very little changes in relative neighborhood of neuronal processes. These findings demonstrate the existence of several constraints in patterning the nervous system and suggest that major substrates for evolutionary novelty lie in the alterations of dendritic structures and synaptic connectivity.

Summary:

Introduction: Comparative studies on nervous system anatomy have a long tradition of offering fundamental insights into the evolution of nervous systems and, consequently, the evolution o behavior. Traditionally, such comparative studies have relied on relatively coarse anatomical and morphometric comparisons. The relative simplicity of nematode nervous systems facilitates the determination and subsequent comparison of neuroanatomical features of distinct nemotode species, thereby enabling an understanding of how members of the same phylum, sharing a common body plan, can engage in very distinct behaviors.

Results: The amphid sensory circuitry of P. pacificus is characterized by a combination of 3D reconstructions from TEM sections of two young adult hermaphrodites, as well as live dye uptake and transgene reporter analysis. The amphibian amphid neuronal network is similar to that of C. elegans. The cell bodies of all amphid neurons and the amphid sheath cell (AMsh) are located in the lateral ganglion posterior to the nerve ring.

Discussion: In the absence of morphological characteristics such as the winged endings of AWx neurons in C. elegans we proposed neuronal homologies between P. pacificus and C.elegan aphid neurons based on their relative cell body positions, axon projections, the manner of dendrite entry into the amphid sheath cell, the number of cilia in channel, dye filling properties, and the connections to first layer interneurons.

Figures:

Figure 1: EM of the amphid sensillum and two other sensory neurons of P. pacificus hermaphrodite adults.

Figure 2: URX is ciliated.

Figure 3: URY, URX, URA and BAG endings.

Figure 4: A comparison of select non-amphid ciliated neurons in three nematode species.

Figure 5: Video of URX 3D reconstruction.

Figure 6: Video of URY 3D reconstruction.

Figure 7: Video of URA 3D reconstruction.

Figure 8: Position of amphid cilia in AMso and AMsh.

Figure 9: Posterior-anterior path of amphid dendrite entry into the left sheath glia of specimen 107.

Figure 10: Posterior-anterior path of amphid dendrite entry into the right sheath glia of specimen 107.

Figure 11: Overview of the amphid sensilla.

Figure 12: 3D rendering of AM1(ASH) neurons.

Figure 13: 3D rendering of AM2(ADL) neurons.

Figure 14: 3D rendering of AM3(AWA) neurons.

Mentioned biomedical entities:

Genes: ADF, ADL, AM11, AM5, AM6, ASE, ASER, ASK, B, BAG, CHE-1, DAF-6, FLP, FP, GA, PHB, PPA01143, Ppa-nhr-66, Ppa-rpl-23, URB, VG, gcy, gcy-6, lsy-6, odr-7, str-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, epidermal cell, neuron, neurotransmitter, secretory cell, sheath cell

Organisms: A. complexus, Acrobeles complexus, C. briggsae, C. elegans, Drosophila melanogaster, E. coli, H. contortus, Haemonchus, Haemonchus contortus, Homo sapiens, Oesophagostomum dentatum, Onchocerca volvulus, Rhabditidae, swine

Tissues: cuticle, dorsal, neuron, tubular, ventral

Pathways: N/A

Query Result 10/25

HRPK-1, a conserved KH-domain protein, modulates microRNA activity during Caenorhabditis elegans development

Li, Li; Veksler-Lublinsky, Isana; Zinovyeva, Anna; Miska, Eric A

PLoS Genetics

Year: 2019 PMID: 31584932 Citations: 12 Score: 13.69 DOI: 10.1371/journal.pgen.1008067

Keywords Hits: lsy-6: 28, che-1: 0

Keywords Frequency (TF): lsy-6: 0.004; che-1: 0.000

Abstract:
microRNAs (miRNAs) are potent regulators of gene expression that function in diverse developmental and physiological processes. Argonaute proteins loaded with miRNAs form the miRNA Induced Silencing Complexes (miRISCs) that repress gene expression at the post-transcriptional level. miRISCs target genes through partial sequence complementarity between the miRNA and the target mRNAs 3 UTR. In addition to being targeted by miRNAs, these mRNAs are also extensively regulated by RNA-binding proteins (RBPs) through RNA processing, transport, stability, and translation regulation. While the degree to which RBPs and miRISCs interact to regulate gene expression is likely extensive, we have only begun to unravel the mechanisms of this functional cooperation. An RNAi-based screen of putative ALG-1 Argonaute interactors has identified a role for a conserved RNA binding protein, HRPK-1, in modulating miRNA activity during C. elegans development. Here, we report the physical and genetic interaction between HRPK-1 and ALG-1/miRNAs. Specifically, we report the genetic and molecular characterizations of hrpk-1 and its role in C. elegans development and miRNA-mediated target repression. We show that loss of hrpk-1 causes numerous developmental defects and enhances the mutant phenotypes associated with reduction of miRNA activity, including those of lsy-6, mir-35-family, and let-7-family miRNAs. In addition to hrpk-1 genetic interaction with these miRNA families, hrpk-1 is required for efficient regulation of lsy-6 target cog-1. We report that hrpk-1 plays a role in processing of some but not all miRNAs and is not required for ALG-1/AIN-1 miRISC assembly. We suggest that HRPK-1 may functionally interact with miRNAs by both affecting miRNA processing and by enhancing miRNA/miRISC gene regulatory activity and present models for its activity.

Summary:
Introduction: Robust regulation of gene expression is essential for normal development and cellular homeostasis. microRNAs (miRNAs), small non-coding RNAs 22nt in length, negatively regulate gene expression at the post-transcriptional level. miRNAs can act as developmental switches or can fine tune the expression of the target genes (for review, see ,). Processed miRNA are loaded into their main protein cofactor, Argonaute (AGO), which then associates with members of the GW182 family of proteins, forming the microRNA Induced Silencing Complex (misC). Mature miRISCs bind to the target messenger RNA(s) and repress their translation and/or destabilize the target mRNA ,.

Results: To identify and characterize potential physical and functional interactions of HRPK-1 with miRISC components, we first generated a null allele inhrpk-1. A previously existing deletion allele ofhrk-1,hprk-1) causes an in-frame deletion within thehrmk-1gene, resulting in production of a truncated HRPK-1) protein. Using CRISPR/Cas9-based genome editing, we generated two independent nullalleles ofhk-1 (zen15) andhk-1). Both alleles nearly completely or completely delete thehk1 locus and produce no HRPK protein. Since both null alleles produce a similar loss-of-function phenotype, we have designated the larger deletion,hkk-1(zen17), as the reference null alele forhrPK-1and used it in subsequent analyses.

Discussion: HRPK-1 is a protein that interacts with miRNA pathways and is required for efficient activity of several miRNAs and miRNA families. The synergistic genetic interaction betweenhrpk-1mutations andmir-35-41(nDf50)is also consistent with a hypothesized role forhrk-1as a positive miRNA co-factor. This work was supported by the National Institutes of Health (grant HL01-0560) and the National Cancer Institute (granted HG01-0040).

Figures:
Figure 1: Multiple hrpk-1 alleles produce deletions within the hrpk-1 locus; ALG-1 co-precipitates with HRPK-1.

Figure 2: hrpk-1 mutations cause developmental defects at 20C and 25C.

Figure 3: hprk-1 functionally interacts with the let-7 family of miRNAs.

Figure 4: hrpk-1 function is required for efficient lsy-6 miRNA activity.

Figure 5: hrpk-1 mutations enhance mir-35-41(nDf51) mutant phenotype at 20C.

Figure 6: Endogenously tagged hrpk-1 :: gfp transgene (hrpk-1(zen64)) is expressed throughout C . elegans development.

Figure 7: miRNA abundances and miRISC formation in hrpk-1 mutant animals.

Figure 8: Models representing potential mechanisms through which HRPK-1 may affect miRNA activity.

Mentioned biomedical entities:
Genes: AGO, AIN-1, AIN-2, ALG-1, ALG-2, ASER, Argonaute, CGH-1, GLD-1, Hui, IP, NEB, NHL-2, P40, PAM, PLD, PLK1, Tubulin, VGLN-1, Vigilin,

cog-1, hnRNPA1, hnRNPK, let-7, lsy-6, miR-84, mir-241, mir-84, tubulin, vgln-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, seam cell

Organisms: human

Tissues: gonad, tube

Pathways: N/A

Query Result 11/25

Ammonium-Acetate Is Sensed by Gustatory and Olfactory Neurons in Caenorhabditis elegans, bNH4Ac Attracts C. elegans.

Frkjr-Jensen, Christian; Ailion, Michael; Lockery, Shawn R.; Hart, Anne C.

PLoS ONE

Year: 2008 PMID: 18560547 Citations: 18 Score: 12.57 DOI: 10.1371/journal.pone.0002467

Keywords Hits: lsy-6: 0, che-1: 27

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.006

Abstract:

, bCaenorhabditis elegans chemosensation has been successfully studied using behavioral assays that treat detection of volatile and water soluble chemicals as separate senses, analogous to smell and taste. However, considerable ambiguity has been associated with the attractive properties of the compound ammonium-acetate (NH4Ac). NH4Ac has been used in behavioral assays both as a chemosensory neutral compound and as an attractant.

Summary:

Background: Caenorhabditis elegans chemosensation has been successfully studied using behavioral assays that treat detection of volatile and water soluble chemicals as separate senses, analogous to smell and taste. However, considerable ambiguity has been associated with the attractive properties of the compound ammonium-acetate (NH 4 Ac). NH 4 AC has been used in behavioral astests both as a chemosensory neutral compound and as an attractant.

Introduction: Chemical compounds that are attractive toC. eleganshave been classified in several different kinds of behavioral assays. Wardassayed water soluble chemoattraction in radial gradients of attractant. Attraction to anions (Cl, Br, I) and cations (Na, Li, K, Mg2) are attractive when peak gradient concentrations are 220 mM. Similar results were seen in an alternative assay in which worms choose between two streams of liquid containing different attractants.

Results: Ammonia and acetic acid are volatile attractants. We hypothesized that NH4Ac could be detected by an odorant pathway. We assayed attraction to acetylcholine and ammonium separately by spotting the attractants on the lid. Both compounds were attractive Thus, C. eleganscan sense ammonia or acetoylcholine as distinct attractants, and thus NH 4Ac can be detected.

Discussion: Ammonium-acetate is an attractive odorant in both water soluble and odorants chemotaxis assays. Ammoniac is used at high concentrations in the odors as well as in the water sorbent assay. It is unlikely that the odourant properties of NH4Ac affect the response ofC. elegansin the standard chemovaxis test developed by Bargmann and colleagues.

Figures:

Figure 1: Odorant chemotaxis to NH 4 Ac.

Figure 2: Genetic analysis of chemotaxis to NH 4 Ac presented in water soluble or odorant form.

Figure 3: Genetic analysis of ammonium and acetate signal transduction pathways.

Figure 4: Genetic analysis of the relative ionic contributionsto water soluble chemotaxis assays.

Mentioned biomedical entities:

Genes: ADF, ASE, ASK, B, BAG, C, CEH-36, DAF-11, GPA-3, MA, MO, ODR-1, ODR-3, OSM-9, Salt, TAX-2, ceh-36, che-3, odr-1, odr-10, odr-7, tax-4

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: Escherichia coli

Tissues: N/A

Pathways: N/A

Query Result 12/25

A Toolkit and Robust Pipeline for the Generation of Fosmid-Based Reporter Genes in C. elegans, bFosmid Recombineering

Tursun, Baris; Cochella, Luisa; Carrera, Ins; Hobert, Oliver; Hart, Anne C.

PLoS ONE

Year: 2009 PMID: 19259264 Citations: 124 Score: 12.39 DOI: 10.1371/journal.pone.0004625

Keywords Hits: lsy-6: 0, che-1: 4

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.001

Abstract:

Engineering fluorescent proteins into large genomic clones, contained within BACs or fosmid vectors, is a tool to visualize and study spatiotemporal gene expression patterns in transgenic animals. Because these reporters cover large genomic regions, they most likely capture all cis-regulatory information and can therefore be expected to recapitulate all aspects of endogenous gene expression. Inserting tags at the target gene locus contained within genomic clones by homologous recombination (recombineering) represents the most straightforward method to generate these reporters. In this methodology paper, we describe a simple and robust pipeline for recombineering of fosmids, which we apply to generate reporter constructs in the nematode C. elegans, whose genome is almost entirely covered in an available fosmid library. We have generated a toolkit that allows for insertion of fluorescent proteins (GFP, YFP, CFP, VENUS, mCherry) and affinity tags at specific target sites within fosmid clones in a virtually seamless manner. Our new pipeline is less complex and, in our hands, works more robustly than previously described recombineering strategies to generate reporter fusions for C. elegans expression studies. Furthermore, our toolkit provides a novel recombineering cassette which inserts a SL2-spliced intercistronic region between the gene of interest and the fluorescent protein, thus creating a reporter controlled by all 5 and 3 cis-acting regulatory elements of the examined gene without the direct translational fusion between the two. With this configuration, the onset of expression and tissue specificity of secreted, sub-cellular compartmentalized or short-lived gene products can be easily detected. We describe other applications of fosmid recombineering as well. The simplicity, speed and robustness of the recombineering pipeline described here should prompt the routine use of this strategy for expression studies in C. elegans.

Summary:

Introduction: The expression pattern of a gene is essential for the complete understanding of its function. In essentially every experimental system including our system of choice, the nematodeC. elegans- the ultimate product of gene expression can be directly visualized in the form of a fluorescent-protein reporter, such as GFP, attached to the gene of interest. A key issue in generating reporter gene fusions is that, in order to accurately reflect all aspects of the temporal and spatial pattern of expression of the gene, as manycis-regulatory control regions as possible need to be included in the DNA construct. However, quite often, expression patterns are inferred from promoter fusion in which only a few kilobases of sequence upstream of the predicted start of the Gene. However it has become increasingly clear that regulatory information that affects the expression pattern a Gene can be found in every part of the respective locus, such introns or downstream of a gene (e.g.

Results and Discussion: Recombineering of fosmids to create a reporter construct in which we recombineered a C. elegansoptimized version of mChOpti with a hidden, intronic scar into theche-1locus, which encodes a Zn-finger transcription factor. We describe here some selected, representative results obtained with individual cassettes. We recombinered gfp,cfpandyfpwith ahidden, intrononic scar in thelsy-2locus. Each recompineered fosmidimulates thelfp mutant phenotype and results in broad expression in all tissue types.

Conclusion: A recombineering pipeline that enables the rapid engineering of reporter gene constructs. Our toolkit offers recombinering cassettes for a variety of tagging options that do not affect the functionality of the engineered gene. Our procedure is robust, time- and cost-efficient and easy to adopt as a standard technique for gene expression analysis in C. elegans. In addition, the overall approach can be easily adapted for a number of other applications.

Figures:

Figure 1: Timeline comparison of available recombineering protocols.

Figure 2: Recombineering pipeline.

Figure 3: A bicistronic recombineering cassette.

Figure 4: Double recombineering.

Figure 5: lsy-2 reporter generated by fosmid recombineering.

Figure 6: che-1 reporter generated by fosmid recombineering.

Figure 7: snb-1 reporter generated by bicistronic fosmid recombineering.

Figure 8: Additional applications of fosmid recombineering.

Mentioned biomedical entities:
Genes: ASE, ASER, B, CFP, CHE-1, FPs, FgF, Flp, GRASP, SNB-1, cfp, cog-1, galK, snb-1, unc-119

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: E. coli

Tissues: Larval, tissue

Pathways: N/A

Query Result 13/25

Functional identification of microRNA-centered complexes in C. elegans

Hebbar, Shilpa; Panzade, Ganesh; Vashisht, Ajay A.; Wohlschlegel, James A.; Veksler-Lublinsky, Isana; Zinovyeva, Anna Y.

Scientific Reports

Year: 2022 PMID: 35504914 Citations: 2 Score: 12.16 DOI: 10.1038/s41598-022-10771-2

Keywords Hits: lsy-6: 41, che-1: 0

Keywords Frequency (TF): lsy-6: 0.005; che-1: 0.000

Abstract:

microRNAs (miRNAs) are crucial for normal development and physiology. To identify factors that might coordinate with miRNAs to regulate gene expression, we used 2O-methylated oligonucleotides to precipitate Caenorhabditis elegans let-7, miR-58, and miR-2 miRNAs and the associated proteins. A total of 211 proteins were identified through mass-spectrometry analysis of miRNA co-precipitates, which included previously identified interactors of key miRNA pathway components. Gene ontology analysis of the identified interactors revealed an enrichment for RNA binding proteins, suggesting that we captured proteins that may be involved in mRNA lifecycle. To determine which miRNA interactors are important for miRNA activity, we used RNAi to deplete putative miRNA co-factors in animals with compromised miRNA activity and looked for alterations of the miRNA mutant phenotypes. Depletion of 25 of 39 tested genes modified the miRNA mutant phenotypes in three sensitized backgrounds. Modulators of miRNA phenotypes ranged from RNA binding proteins RBD-1 and CEY-1 to metabolic factors such as DLST-1 and ECH-5, among others. The observed functional interactions suggest widespread coordination of these proteins with miRNAs to ultimately regulate gene expression. This study provides a foundation for future investigations aimed at deciphering the molecular mechanisms of miRNA-mediated gene regulation.

Summary:

Introduction: Developmental and physiological processes require precise spatio-temporal regulation of gene expression. One post-transcriptional gene regulatory mechanism is directed by a class of small non-coding RNAs called miRNAs. miRNA regulate a wide range of developmental and cellular processes, with dysregulated miRNA activity prevalent in diseases. miRISC binding to the target mRNA via partial sequence complementarity between a miRNA and 3 UTR of mRN triggers a series of gene silencing mechanisms including translation inhibition, decapping, and mRAN decay.

Results: Identifying proteins associated with let-7, miR-58, and miR-2 miRNAs using a shotgun proteomics approach. Identification of proteins associated in the complexes with these miRNA. Identification and identification of proteins that may regulate miRNA activity. Identification, identification, and identification. of proteins enriched in miRNA co-precipitites. Identification. of protein complexed with let-77, mi-58 and mi-2-2 miRNA of interest. Identification

Discussion: Proteomics analysis of miRNA-centered complexes reveals functional interaction between let-7 and interacting factors. Knockdown of six genes significantly modulated the vulval bursting phenotype oflet-7(n2853)mutants. Five of these interactors were identified in let-7 PD experiments. Knokdown of 16 putative interactor proteins did not modify the resulting vulvular bursting defect. These results suggest that let-7 interactor factors indeed functionally coordinate with let-7 activity.

Figures:

Figure 1: miRNA pulldowns (PDs) identify putative physical interactors of miRNA complexes. (a) Protein complexes associated with miRNAs were isolated using 2-OMe modified oligonucleotides complementary to the miRNA of interest and subjected to MudPIT mass spectrometry analysis (drawn using BioRender (biorender.com)). (b d) Average Normalized Spectral Abundance Factor (NSAF) values of factors identified in pulldown experiments (Y-axis) are plotted against that of corresponding scrambled oligo controls (X-axis) for (b) let-7 PD, (c) miR-58 PD and (d) miR-2 PDs. Highlighted in black are key miRISC components and components of miRNA biogenesis machinery. (e) Venn diagram showing the number of factors that passed a set of criteria to qualify as a putative interactor (for description of criteria, see Methods). Percentages shown here are percentage of total number of proteins (308) that passed the criteria in interaction datasets including ALG-1 19 . Proteins commonly co-precipitated with ALG-1, let-7, miR-58, and miR-2, ALG-1 IP, let-7 PD, and miR-58 PD, let-7 PD, miR-58 PD, miR-2 PD, and ALG-1 IP, miR-58 PD, and miR-2 PD. (f) Factors commonly identified in two or more interaction datasets include core miRNA machinery components. Symbols in the left column correspond to overlapping protein populations in (e). Colored boxes show which IP or PD precipitated the proteins listed in the left column. (g) NSAF values of proteins identified in pulldowns with let-7 complementary oligonucleotide from mir-48 mir-241(nDf51); mir-84(n4037) mutant animals, plotted against NSAF of proteins identified in scrambled oligo control. (h) let-7 complementary oligonucleotide precipitates a partially overlapping set of factors from let-7 -family miRNA deletion backgrounds.

Figure 2: miRNA pulldowns identify components of translation machinery and mRNA processing factors, which may form a functional network. (a d) Terms identified through GO analysis for biological processes (top four), cellular components (top four) and molecular function categories (top three) among let-7 PDs in (a) wild type, (b) mir-48 mir-241(nDf51); mir-84(n4037) backgrounds, (c) miR-58 PD and (d) ALG-1 IP 19 . Total number of proteins classified under each term shown adjacent to respective bars. (eh) RNA binding domains identified in factors captured in let-7 PDs from (e)

wild type, and (f) mir-48 mir-241(nDf51); mir-84(n4037)) backgrounds, (g) miR-58 PD and (h) ALG-1 IP. The FDR adjusted p-values for enrichment of proteins harboring individual domains are shown within the respective bars. (i l) The reproducibly enriched proteins form functional network. Network analysis was performed using STRING 28 on top 40 enriched interactors in (i) let-7 PD (wild type), (j) let-7 PD (mir-48 mir-241(nDf51); mir-84(n4037)) (k) miR-58 PD and (l) ALG-1 IP. Ribosomal proteins were excluded from this analysis. Thickness of edges represents degree of confidence of functional linkages. The number of edges is significantly higher than expected, with a p -value1.0e16.

Figure 3: RNAi knockdown of genes encoding ALG-1 and let-7-associated factors genetically modifies let-7(n2853) vulval bursting phenotype. (a) let-7(n2853) mutants show partially penetrant vulval bursting phenotype at 15C. RNAi of (b) genes of previously reported conserved interactors of AGO 19 and (c) let-7-precipitated factors in wild type and let-7(n2853) mutants. Each dot represents an independent RNAi experiment. Statistical significance was determined by one-way ANOVA with post hoc Bonferroni correction. (d) Effects of vector and rbd-1 RNAi on pdpy-30::gfp::lin-41 3UTR reporter expression in the vulval cells of wild type and let-7(n2853) L4 larvae. Compromised miRNA activity in let-7(n2853) background leads to de-repression of pdpy-30::gfp::lin-41 3 UTR reporter expression levels (quantified in e). (e) RNAi of genes of modifiers of let-7(n2853) vulval bursting phenotype affects pdpy-30::gfp::lin-41 3UTR reporter expression levels in let-7(n2853) background. Each dot represents the relative intensity in an individual worm. Vulval precursor cells used for fluorescence quantification are highlighted with dashed circles. T-test was used to determine statistical significance. Vector=empty vector RNAi control.

Figure 4: RNAi knockdown of genes encoding for putative ALG-1 and miRNA interactors modulates mir-48 mir-241(nDf51) seam cell lineage and hypodermal cell fate defects. (a) A schematic representation of seam cell lineages of wild type and mir-48 mir-241(nDf51) mutant animals throughout C. elegans larval development. (b) Wild type young adult animals express adult cell fate marker, col-19::gfp, in seam and hypodermal cells, while mir-48 mir-241(nDf51) mutants show lack the col-19::gfp expression in the hypodermis as young adults some of the time (quantified in c). (c , d) Effects of RNAi knockdown of genes of (c) conserved AGO 19 and (d) other miRISC interactors on hypodermal col-19::gfp expression in wild type and mir-48 mir-241(nDf51) mutants. Each dot represents an independent RNAi experiment. (e , f) Effects of RNAi knockdown of genes of (e) conserved AGO 19 and (f) other miRISC interactors on the seam cell number in mir-48 mir-241(nDf51) mutant young adults . Each dot represents seam cell number in an individual worm. Statistical significance was determined by One-way ANOVA with post hoc Bonferroni correction. (g) Expression of hbl-1::gfp::hbl-1 3 UTR reporter in control and pdi-2 RNAi in wild type and mir-48 mir-241(nDf51) L3 larvae. Arrows indicate hypodermal cells and arrowheads indicate seam cells. (h) RNAi of most genes that genetically modified heterochronic defects in mir-48 mir-241(nDf51) mutants significantly affected hbl-1::gfp::hbl-1 3 UTR reporter expression in both wildtype and mir-48 mir-241(nDf51) backgrounds. Each dot represents an independent RNAi experiment. T-test was used to determine statistical significance . Vector=empty vector RNAi control.

Figure 5: ALG-1 and miRNA interactors genetically interact with lsy-6(ot150) . (a) lsy-6 determines the asymmetric expression of chemoreceptors in ASER and ASEL neurons, with plim-6::gfp expression marking ASEL neuronal cell fate. ASEL cell defective phenotype in the lsy-6(ot150) animals can be observed by lack of plim-6::gfp expression some of the time (quantified in b). RNAi knockdown of (b) conserved AGO 19 and (c) miRNA/ALG-1 interactors in plim-6::gfp or lsy-6(ot150); plim-6::gfp animals . Each dot represents an independent RNAi experiment. Statistical significance was determined by One-way ANOVA with post hoc Bonferroni correction. (d) cog-1 expression in uterine cells is repressed by pcog-1::lsy-6 . RNAi of F33D11.10 depresses cog-1::gfp in that background (quantified in e). (e) RNAi of genes that modify lsy-6(ot150) phenotype alleviates lsy-6-mediated repression of cog-1::gfp in uterine cells. Each dot represents an independent RNAi experiment. T-test was used to determine statistical significance. Vector=empty vector RNAi control.

Figure 6: Summary of functionally relevant interactors and models for possible mechanisms through which miRNA and ALG-1 interactors affect gene regulation. (a) Venn diagram showing the overlap of functional hits among the three functional assays. (b d) Percentages of assayed factors that modulated phenotypes upon their respective gene knockdowns in (b) let-7(n2853) , (c) mir-48 mir-241(nDf51) , and (d) lsy-6(ot150) assays, with protein physical interaction status shown on the x-axis. One ALG-1 interactor was also identified in miR-2 pulldown in addition to ALG-1 IP. The number of genes that modulated miRNA phenotypes over total number of genes tested is shown above respective bars. (e f) RNA binding proteins found interacting with ALG-1 as well as miRNAs could be involved in (e) miRNA duplex processing and/or (f) facilitating miRISC activity on the targets by associating with downstream effectors. (g) Metabolic enzymes with putative RNA binding ability, such as ECH-5, could bind target mRNAs and promote miRISC activity by recruiting deadenylases and decapping proteins. Panels (e , f), and (g) were drawn using BioRender (https://biorender.com).

Mentioned biomedical entities:

Genes: 8, AGO, AIN-1, AIN-2, ALG-1, ALG-2, AUH, Ago2, Argonaute, CGH-1, Dicer, IFG-1, LET-363, NHL-2, PAB-1, RBFOX3, RBP, SMG-2, Staufen, VNC, cog-1, eIF4G, ifg-1, larp-1, let-363, let-7, lin-41, lsy-6, mTOR, miR-2, miR-241, miR-84, mir-241, mir-84, rbd-1, rnp-7

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, seam cell

Organisms: Caenorhabditis, E. coli, human, humans, mice, mouse

Tissues: hypodermis, tissue

Pathways: N/A

Query Result 14/25

LIN-42, the Caenorhabditis elegans PERIOD homolog, Negatively Regulates MicroRNA Transcription

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PLoS Genetics

Year: 2014 PMID: 25032706 Citations: 29 Score: 12.02 DOI: 10.1371/journal.pgen.1004486

Keywords Hits: lsy-6: 21, che-1: 0

Keywords Frequency (TF): lsy-6: 0.003; che-1: 0.000

Abstract:

During C. elegans development, microRNAs (miRNAs) function as molecular switches that define temporal gene expression and cell lineage patterns in a dosage-dependent manner. It is critical, therefore, that the expression of miRNAs be tightly regulated so that target mRNA expression is properly controlled. The molecular mechanisms that function to optimize or control miRNA levels during development are unknown. Here we find that mutations in lin-42, the C. elegans homolog of the circadian-related period gene, suppress multiple dosage-dependent miRNA phenotypes including those involved in developmental timing and neuronal cell fate determination. Analysis of mature miRNA levels in lin-42 mutants indicates that lin-42 functions to attenuate miRNA expression. Through the analysis of transcriptional reporters, we show that the upstream cis-acting regulatory regions of several miRNA genes are sufficient to promote highly dynamic transcription that is coupled to the molting cycles of post-embryonic development. Immunoprecipitation of LIN-42 complexes indicates that LIN-42 binds the putative cis-regulatory regions of both non-coding and protein-coding genes and likely plays a role in regulating their transcription. Consistent with this hypothesis, analysis of miRNA transcriptional reporters in lin-42 mutants indicates that lin-42 regulates miRNA transcription. Surprisingly, strong loss-of-function mutations in lin-42 do not abolish the oscillatory expression patterns of lin-4 and let-7 transcription but lead to increased expression of these genes. We propose that lin-42 functions to negatively regulate the transcriptional output of multiple miRNAs and mRNAs and therefore coordinates the expression levels of genes that dictate temporal cell fate with other regulatory programs that promote rhythmic gene expression.

Summary:

Introduction: MicroRNAs (miRNAs) are non-coding RNA molecules that post-transcriptionally regulate gene expression. The maturation of miRNAs is a stepwise process that begins with the RNA polymerase II-dependent transcription of long capped and polyadenylated primary miRNA. Most pri-miRNAs are then endonucleolytically cleaved by the nuclear Microprocessor complex, composed of Drosha (an RNase III enzyme) and its binding partner Pasha, to yield a 70 nt precursor miRNA hairpin (pre-mrna). After export to the cytoplasm, the pre-mimrn is cleaded by Dicer (a second Type III RNase) yielding a 22 n t duplex that consists of the mature miRNA and its corresponding passenger RNA. The mature single-stranded 22 nd miRNA is then loaded into the Argonaute and GW182 to form the miRNA-induced Silencing Complex (misC). Through partial complementary base-pairing between the miRNa and target mRNA, the miRISC complex negatively regulates gene expression by either translational repression or mrRNA degradation,. In vivo, target nRNA down-regulation is directly proportional to the amount of miRN associated with miRIS C.

Results: lin-4(ma161) and alg-1(mu192) mutations correct aberrant cell fate transitions mediated by miRNAs. These mutations also alter the ability of processed miRNA to repress downstream target mRNAs and re-express L1-specific seam cell division patterns. These mutants also fail to express adult-specific gene regulatory programs. These findings provide a novel genetic context for studying the heterochronic pathway.

Discussion: Using an unbiased genetic approach, we sought to identify factors that modulate the expression of miRNAs that are critical for controlling temporal patterns of development throughout post-embryonic development. These efforts identifiedlin-42, theC. eleganshomolog of the circadianperiodgene, as a component that not only modulates heterochronic miRNA expression, but also regulates the expression a wide range of broadly expressed, and functionally distinct,C.

Figures:

Figure 1: Mutations in lin-42 suppress defects in temporal gene expression and cell lineage phenotypes of heterochronic miRNA mutants.

Figure 2: lin-42 mutations lead to the overexpression of several miRNAs.

Figure 3: lin-42 suppresses neuronal phenotypes associated with lsy-6 miRNA-mediated cell fate specification.

Figure 4: The promoters of three different miRNAs display dynamic expression patterns that are coupled to the larval molting cycle.

Figure 5: LIN-42 binds to the putative regulatory regions of both miRNAs and mRNAs.

Figure 6: lin-42 controls the output of lin-4 and let-7 transcription.

Figure 7: A model for lin-42 function in regulating post-embryonic miRNA expression.

Mentioned biomedical entities:

Genes: ALG-1, ALG-2, ASE, ASER, Argonaute, COG-1, Dicer, LIN-42, PRG-1, TF, alg-1, cog-1, col-12, csh4, eL1, hbl-1, hyp7, let-7, lin-4, lsy-6, miR-84, tRNAGly

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, seam cell

Organisms: human

Tissues: cuticle, hypodermis

Pathways: N/A

Query Result 15/25

Poly(A)-binding proteins are required for microRNA-mediated silencing and to promote target deadenylation in C. elegans

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Nucleic Acids Research

Year: 2016 PMID: 27095199 Citations: 23 Score: 11.59 DOI: 10.1093/nar/gkw276

Keywords Hits: lsy-6: 15, che-1: 0

Keywords Frequency (TF): lsy-6: 0.003; che-1: 0.000

Abstract:

Cytoplasmic poly(A)-binding proteins (PABPs) link mRNA 3 termini to translation initiation factors, but they also play key roles in mRNA regulation and decay. Reports from mice, zebrafish and Drosophila further involved PABPs in microRNA (miRNA)-mediated silencing, but through seemingly distinct mechanisms. Here, we implicate the two Caenorhabditis elegans PABPs (PAB-1 and PAB-2) in miRNA-mediated silencing, and elucidate their mechanisms of action using concerted genetics, protein interaction analyses, and cell-free assays. We find that C. elegans PABPs are required for miRNA-mediated silencing in embryonic and larval developmental stages, where they act through a multi-faceted mechanism. Depletion of PAB-1 and PAB-2 results in loss of both poly(A)-dependent and -independent translational silencing. PABPs accelerate miRNA-mediated deadenylation, but this contribution can be modulated by 3UTR sequences. While greater distances with the poly(A) tail exacerbate dependency on PABP for deadenylation, more potent miRNA-binding sites partially suppress this effect. Our results refine the roles of PABPs in miRNA-mediated silencing and support a model wherein they enable miRNA-binding sites by looping the 3UTR poly(A) tail to the bound miRISC and deadenylase.

Summary:

INTRODUCTION: GW182 proteins bridge miRNAs to effector silencing machineries. In mammals and flies, the GW 182 proteins encode an Argonaute-binding domain characterized by GW/WG repeats and a silence domain (SD) which interacts with CNOT1, a scaffold protein of the multi-subunit CCR4/NOT1 deadenylase complex (,). As such, the miRISC functional architecture revolves around the core Argonautes and GW184 proteins.

RESULTS: PAB-1 and PAB-2 associate with miRISC in C. elegans embryos. We previously conducted a comparative proteomic analysis of miRIS captures using pull-down of 2-O-methylated (2-O-Me) oligonucleotide mimics of single target sites for maternal and zygotic miRNA families highly enriched in the embryo. As single miRNA-binding sites do not trigger target deadenylation we reasoned that capture based on mimics for potent combinations of target sites would better reflect components of active miRIS C.

DISCUSSION: In this study, multi-pronged genetic and biochemistry approaches indicate that PAB-1 and PAB-2, the C. elegans orthologs of human cytoplasmic PABPC1, interact physically and functionally with miRISC. PABs function in miRNA-mediated silencing was assessed not only using artificial miRNA reporters, but by examining the functional outcome of regulation on endogenous targets, embedded in two phenocritical genetic cascades. Probing mechanistic bases using embryonic cell-free assays uncovered a multi-faceted mechanism: depletion of PABPs beyond detectable levels prevented repression of polyadenylated and un-adenilated mRNA reporters , yet still permitted miRNA mediated deadenylation. Furthermore, we show that 3UTR-specific features, such as miRNA binding site potency and their distance with the poly(A) tail, can modulate the contribution of PAB-s in target deadenilation.

Figures:

Figure 1: PAB-1 and PAB-2 interact with miRISC in Caenorhabditis elegans embryos. (A) ALG-1/2 western blot analysis of 2- O -Me pull-downs. 5 Biotinylated 2- O -Me oligos containing miR-35, CeBantam or both sites were used to pull down the miRISC WT strain (N2) embryonic extract. ALG-1 and ALG-2 were detected using near-infrared fluorescent western blot (LiCOR). (B) Proteins identified in MuDPIT analyses of a dual site (CeBantam + miR-35) 2- O -Me pull-down. Only proteins which were detected in two independent purifications and were absent from pull-downs with mutated binding sites are included here. For each protein, the number of peptides (pep) found in each experiment and the coverage (%cov) of these peptides for the full-length protein are indicated. (C) Venn diagram comparing detected interactions with the dual site (CeBantam + miR-35) pull-down to single-site RISC pull-down, and AIN-2-GFP IP. (D) A GST pull-down using either GST or GST-PAIP2 was performed either on WT (N2) or mutant pab-2(0) (ok1851) embryonic extract in the presence of 0.1 ng/l RNAse A. The bound proteins were analyzed on SDS-PAGE and detected by western blot.: non-specific band.

Figure 2: pab-1 and pab-2 genetically cooperate with let-7 and lsy-6 miRNAs. (A) pab-1 and pab-2 genetically interact with the let-7 miRNA. let-7(n2853) animals (L1) were fed with bacterially-expressed dsRNA against the indicated gene, or L4 animals injected with dsRNA (B) and F1 animals were scored for bursting vulva phenotype at the permissive temperature (16C). Results shown are representative of at least two experimental and biological replicates. (C) pab-2 genetically interacts with lsy-6. pab-2(0)(ok1851) were crossed with OH3646 (otIs114(Plim-6::GFP, rol-6(d));

lsy-6(ot150) . Animals were scored for expression of GFP in the ASEL.

Figure 3: Biochemical depletion of PAB1 and PAB-2 delays deadenylation. (A) PAB-1 and PAB-2 levels were assessed by western blot in GST- or GST-PAIP2-treated extracts. A fraction of the extract was analyzed by SDS-PAGE followed by western blotting. (B) RL-6-miR-35-p(A) 86 was subjected to an in vitro deadenylation assay in the presence of a miR-35 or miR-1 2- O -Me inhibitor in GST- or GST-PAIP2-treated extracts. RNA was extracted and analyzed by UREA-PAGE. (C) RL-6-miR-52-p(A) 86 was subjected to an in vitro deadenylation assay in GST- or GST-PAIP2-treated extracts. RNA was extracted and analyzed by UREA-PAGE. (D) Human PABC1 restores deadenylation in the Caenorhabditis elegans embryonic extract. PAB-1/2-depleted extract was supplemented with 115 nM of human PABC1 or GST proteins and a radiolabeled RL-6-miR-35-p(A) 86 was subjected to deadenylation and analyzed by UREA-PAGE. T 1/2 is the half deadenylation time (min).

Figure 4: Depletion or impairment of PAB-1 and PAB-2 prevents miRNA-mediated silencing in vitro . The RL-6-miR-35-p(A) 86 reporter was subjected to a translational repression assay in GST- or GST-PAIP2-treated extracts (A) or in presence of 2.5M soluble GST or GST-PAIP2 (B). Translational activity was monitored through measurement of RL activity in the presence of miR-35 2- O -Me inhibitoror a non-cognate miR-1 2- O -Me inhibitor. The RL-6-miR-52-p(A) 86 reporter was subjected to a translational repression assay in GST- or GST-PAIP2-treated extracts (C) or in presence of 5 M soluble GST or GST-PAIP2 (D). Translational activity was monitored through measurement of RL activity in the presence of miR-52 2- O -Me inhibitoror a non-cognate miR-58 2- O -Me inhibitor.

Figure 5: PAB-1 and PAB-2 are required for miRNA-mediated silencing independently of the poly(A) tail. (A) Design for the RNA used in the experiment. A capped transcript containing a Renilla luciferase ORF and six miR-35 binding sites in its 3UTR, but lacking a poly(A) tail. (B) The reporter was incubated in an embryonic translation extract in the presence of 2- O -Me inhibitor (miR-35) or non-specific inhibitor (miR-1). Translation activity was monitored through measurement of RL activity. Repression activity of the RL-6-miR-35-p(A) 0 transcript was assayed in (C)GST- or(D) GST-PAIP2-treated extracts. Translation counts were monitored over time in the presence of miR-35 or miR-1 2- O -Me inhibitors. (E) RL-6-miR-35-p(A)0 was incubated in GST- or GST-PAIP2-treated extracts. RNA was extracted and analyzed by UREA-PAGE.: significance in a one-sided Welch t -test.

Figure 6: 3UTRs modulate PABP contribution in miRNA-mediated deadenylation. (A) Design of the RNAs used in the experiment. Capped transcripts encode the Renilla luciferase ORF and either three or six miR-35-binding sites in its 3UTR, followed by a linker of 32 nt (L32) or 262 nt (L262). (B) RL-6-miR-35-p(A) 86 L32 and L262 were subjected to an in vitro deadenylation assay in GST- or GST-PAIP2-treated extract. (C) RL-3-miR-35-p(A) 86 L32 and L262 were subjected to an in vitro deadenylation assay in GST- or GST-PAIP2-treated extract. (D) The 3UTR-poly(A) loop model of PABP function in miRNA-mediated deadenylation. PABPs enables miRNA-binding sites by facilitating access to the poly(A) tail to miRISC and to theassociated deadenylase. Longer 3UTRs exacerbate, and shorter 3UTRs suppress PABP dependency. T 1/2 correspond to the half deadenylation time (min).

Mentioned biomedical entities:

Genes: AIN-1, AIN-2, ALG-1, ALG-2, Argonaute, CNOT1, GST, GW, HD2, IFG-1, L32, MOP, NEB, PAB, PAB-1, PAB1, PABP, PABPC1, PAIP2, Pabpc1, S10, TBB-2, ain-1, ain-2, bantam, cog-1, let-7, lsy-6, miR-52, pab-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: Human, human, mice, rabbits, zebrafish

Tissues: N/A

Pathways: N/A

Query Result 16/25

Single-cell transcriptional analysis of taste sensory neuron pair in Caenorhabditis elegans

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Nucleic Acids Research

Year: 2010 PMID: 19875417 Citations: 18 Score: 11.43 DOI: 10.1093/nar/gkp868

Keywords Hits: lsy-6: 9, che-1: 17

Keywords Frequency (TF): lsy-6: 0.002; che-1: 0.003

Abstract:

The nervous system is composed of a wide variety of neurons. A description of the transcriptional profiles of each neuron would yield enormous information about the molecular mechanisms that define morphological or functional characteristics. Here we show that RNA isolation from single neurons is feasible by using an optimized mRNA tagging method. This method extracts transcripts in the target cells by co-immunoprecipitation of the complexes of RNA and epitope-tagged poly(A) binding protein expressed specifically in the cells. With this method and genome-wide microarray, we compared the transcriptional profiles of two functionally different neurons in the main C. elegans gustatory neuron class ASE. Eight of the 13 known subtype-specific genes were successfully detected. Additionally, we identified nine novel genes including a receptor guanylyl cyclase, secreted proteins, a TRPC channel and uncharacterized genes conserved among nematodes, suggesting the two neurons are substantially different than previously thought. The expression of these novel genes was controlled by the previously known regulatory network for subtype differentiation. We also describe unique motif organization within individual gene groups classified by the expression patterns in ASE. Our study paves the way to the complete catalog of the expression profiles of individual C. elegans neurons.

Summary:

INTRODUCTION: Each neuron in a mature nervous system has various physiological, biochemical and morphological properties, which are defined by the specific set of expressed genes in the neurons. Therefore, a description of the transcriptional profiles of each neuron would provide a basis for understanding the molecular mechanisms that generate neuronal diversity. With its relatively simple nervous system and fully-sequenced genome, the model nematode Caenorhabditis elegansoffers a unique opportunity to attempt this challenging research objective).

RESULTS: Optimizing the mRNA tagging/microarray profiling method for single cells. To enrich for transcripts from the single mature taste neurons, we generated twoC. eleganslines stably expressing FLAG PABP in ASEL or ASER from chromosomally integrated transgenes. FLAGPABP was expressed by the promoter ofgcy-7orgcy-5, which drives expression exclusively in ASER or ASEL, respectively. Single cell expression of FLAG APBP was confirmed by immunohistochemistry with the anti-FLAG antibody. The efficiency of enrichment was assessed by quantifying the relative amount of the transcripts in the extracted RNA samples by real-time quantitative RT-PCR.

DISCUSSION: This study shows that transcript extraction from single cells is feasible by a modified mRNA tagging method. This technique can be used in applications from single cell to large tissues, so long as an appropriate promoter is available. The major advantages of this method are its independence of any regulatory factors and applicability to every developmental stage, while limitation to generality of this methodology would be the requirement of appropriate cell-specific promoters. Since the transcriptional profiling of a small number of cells in C. eleganswas so far limited to embryonic cells, our study opened the door to the impartial expression profilingof single cells that differentiate or are even generated post-embryonically.

Figures:

Figure 1: FLAG::PABP is expressed in single neurons of the transgenic animals. Expression of FLAG::PABP in the single cells of transgenic worms carrying ASELp::flag::pabp (A) and ASERp::flag::pabp (B) are visualized by immunohistochemistory with anti-FLAG antibodies. Arrowheads point to the immunofluorescence signals. Top panels are immunofluorescence images, and bottom panels are differential interference contrast (DIC) images. Scale bar = 25 m.

Figure 2: Representative fluorescence images for novel ASER-biased genes. (A) Schematic depiction of the locations of ASE neurons. Circles indicate the ASEL and ASER neurons. (BH) Transgenic animals carrying promoter-fused Venus reporter genes. Animals at the L1 stage (BF) and the adult stage (G , H) are shown. Scale bar = 10 m.

Figure 3: Quantification of the expression bias of the newly identified genes in wild-type and mutant backgrounds. (A) Lateralized gene expression in ASE is controlled by the upstream double-negative feedback loop and the downstream transcription factors (6). ASER-biased gcy genes are regulated by lim-6 , whereas hen-1 is independent of lim-6 . Genes in active or inactive state are shown in black or grey, respectively. (B) Quantification of the expression laterality in ASE neurons of the newly identified genes in wild-type and the mutant backgrounds. The expression bias is indicated as follows except for trp-2 : L 0 and 0 R refer to the restricted expression to ASEL or ASER, respectively; L R and L R refer to different levels of expression in ASEL versus ASER; L = R refers to equal expression in ASEL and ASER; 0 = 0 refers to no expression in ASEL nor in ASER. For trp- 2, since strong Venus expression in trp-2p::Venus worms were seen in neurons just medial to ASE, the expression was observed on one side

for each animal. Strong, Weak and None refer to the expression levels at each neuron. The number of examined animals is shown in each graph.

Figure 4: The newly identified genes with ASER-biased expression have ASE motifs. All of the ASER-biased genes had ASE motifs in the promoter region, whereas ASEL-biased ins-32 did not. The location of ASE motifs was shown by arrow heads. Notably, six of the eight ASER-biased genes harbored multiple motifs within the promoter region.

Figure 5: Similar ASE motif organization was observed within each gene set classified by the expression laterality. (A) The occurrence frequency of ASE motifs was high in the genes expressed in ASE neurons. ASEL, Bilateral, ASER and Neither refer to the gene set with corresponding expression in ASE neurons. Genes in the Random set were selected randomly from gene spots in the microarray. (B) Most of the ASEL-biased genes had single ASE motifs, whereas many of the other two gene sets had multiple motifs (P 0.05; randomization test). Bars refer to the average number of motifs. Genes with no ASE motif were discarded, since they were assumed to be indirectly regulated by CHE-1. (C) No strong correlation was observed between the length of a promoter and the number of motifs per promoter. (D) A was most frequently observed at the fourth nucleotide within the ASE motifs of the promoters with ASEL-biased expression, whereas G was the majority in genes expressed in ASER. (E) T was most frequently observed at the seventh nucleotide in ASEL-biased promoters, whereas no strong preference in nucleotide usage was seen in the promoters with equivalent or ASER-biased expression. (F) Expression laterality in ASE neurons of wild-type nlp-5 promoter and mutant promoter that has ASEL-type mutations in ASE motif. Laterality is presented in the same manner as Figure 3 .

Mentioned biomedical entities:
Genes: 27, ASE, ASER, Bad, C, CHE-1, PABP, Rex, T, TRPC, cog-1, die-1, dpy-20, flp, fozi-1, gcy-22, gcy-5, gcy-6, lim-6, lsy-6, nlp-5, nlp-7

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell

Organisms: C. elegans

Tissues: body wall muscle, chemoreceptor, neuron

Pathways: N/A

Query Result 17/25

MicroRNAs: Not Fine-Tuners but Key Regulators of Neuronal Development and Function

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Frontiers in Neurology

Year: 2015 PMID: 26635721 Citations: 42 Score: 8.26 DOI: 10.3389/fneur.2015.00245

Keywords Hits: lsy-6: 7, che-1: 1

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.000

Abstract:

MicroRNAs (miRNAs) are a class of short non-coding RNAs that operate as prominent post-transcriptional regulators of eukaryotic gene expression. miRNAs are abundantly expressed in the brain of most animals and exert diverse roles. The anatomical and functional complexity of the brain requires the precise coordination of multilayered gene regulatory networks. The flexibility, speed, and reversibility of miRNA function provide precise temporal and spatial gene regulatory capabilities that are crucial for the correct functioning of the brain. Studies have shown that the underlying molecular mechanisms controlled by miRNAs in the nervous systems of invertebrate and vertebrate models are remarkably conserved in humans. We endeavor to provide insight into the roles of miRNAs in the nervous systems of these model organisms and discuss how such information may be used to inform regarding diseases of the human brain.

Summary:

Introduction: MicroRNAs (miRNAs) are non-coding RNA molecules with a length of approximately 22 nucleotides, which act as post-transcriptional regulators of gene expression. Discovered just over two decades ago, miRNAs have been found to be abundant in most organisms and critically important for post-transscriptional control of mRNAs by regulating a predicted 60 of protein-coding genes. Prior to the discovery of the miRNA pathway, the lin-14 gene in Caenorhabditis eleganswas shown to be regulated by a 22-nucleototide partially complementary strand of RNA calledlin-4. However, a mechanistic understanding of this process remained unclear until the let-7 gene was shown to encode a complementary sequence oflin-41to regulate developmental timing. This led to a paradigm shift in how mRNR regulation was viewed, and further investigation demonstrated that the miRNR pathway was evolutionarily conserved in most eukaryotes. Since then miRNA have been shown to have been required for key biological processes, such as cell fate, differentiation, apocytosis, and tumor suppression ().

Conclusion: lsy-6 miRNA is a small non-coding RNA that is involved in the nervous system. It is regulated by the lin-4 and let-7 miRNAs. These miRNA are involved in a number of biological processes, including neuronal development, neuron differentiation, and neuron function. These non-coded RNAs are regulated in the brain by the miR-44 and miR-4 miRNA. These RNA are a part of the mRNA family.

Figures:

Figure 1: The miRNA pathway . Primary miRNAs (pri-miRNAs) are transcribed from the genome and form hairpin structures. Nuclear-localized Drosha endonuclease cleaves pri-miRNAs into approximately 70nt precursor miRNAs (pre-miRNAs) which are then transported from the nucleus to the cytoplasm by Exportin-5 (XPO5) via the nuclear pore complex, where they are further cleaved by Dicer into mature 2123nt miRNA fragments. Once the strands separate, the guide strand is loaded into the RISC complex (AGO and different cofactors) to scan the transcriptome for partial complementary target transcripts. These sequences are either repressed by the RISC complex or degraded in P-bodies.

Figure 2: Roles of miRNAs in different stages of neuronal development . miRNAs are involved in the multiple stages of neuronal development in invertebrates and vertebrates. Listed here are the miRNAs we cover in this review that regulate single or multiple stages of neuronal development. cel, Caenorhabditis elegans ; dme, Drosophila melanogaster ; dre, Danio rerio ; hsa, Homo sapiens ; mmu, Mus musculus ; xtr, Xenopus tropicalis .

Mentioned biomedical entities:

Genes: AGO, ASCL1, Akt, Argonaute, CHE-1, COG-1, CamkII, Cxcr4b, DAF-16, Dgcr8, Dicer, EGFR, Emx1, ErbB2, FMR1, FOXO, Fgf, Fgf8, Fmr1, Foxg1, Foxp2, Gsh2, Hes1, LIN-14, LON-2, Let-7, Lmx1b, MAP2, MECP2, MEF-2, MLN51, MYT1L, MeCP2, NEUROD2, NRXN1, Notch, Nrg, Phf6, REST, RhoG, SHANK3, SOX-2, SQV-7, Sox9, Syt1, TLX, TRIM, TRIM32, UNC-40, Wnt, XPO5, ana, chinmo, cnot8, dicer, dme, fgfr1, her5, jigr1, let-7, lin-14, lin-28, lin-4, lsy-6, miR-124, miR-125, miR-125b, miR-137, miR-228, miR-289, miR-8, miR-9, miR-92, miR-92b, miR-9a, mir-124, mir-71, pitx3, zic5

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, glial cell, neural progenitor cell, neural stem cell, neuron, neurotransmitter, stem cell

Organisms: Danio rerio, Drosophila melanogaster, Homo sapiens, Mus musculus, human, humans, mice, mouse, zebrafish

Tissues: epidermis, neuron, synapse, tissue

Pathways: N/A

Query Result 18/25

Cell-type specific sequencing of microRNAs from complex animal tissues

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Nature methods

Year: 2018 PMID: 29481550 Citations: 51 Score: 7.60 DOI: 10.1038/nmeth.4610

Keywords Hits: lsy-6: 5, che-1: 2

Keywords Frequency (TF): lsy-6: 0.001; che-1: 0.000

Abstract:

MicroRNAs (miRNAs) play an essential role in the post-transcriptional regulation of animal development and physiology. However, in vivo studies aimed at linking miRNA-function to the biology of distinct cell types within complex tissues remain challenging, partly because in vivo miRNA-profiling methods lack cellular resolution. We report microRNome by methylation-dependent sequencing (mime-seq), an in vivo enzymatic small RNA-tagging approach that enables high-throughput sequencing of tissue- and cell-type-specific miRNAs in animals. The method combines cell-type-specific 3-terminal 2-O-methylation of animal miRNAs by a genetically encoded, plant-specific methyltransferase (HEN1), with chemoselective small RNA cloning and high-throughput sequencing. We show that mime-seq uncovers the miRNomes of specific cells within C. elegans and Drosophila at unprecedented specificity and sensitivity, enabling miRNA profiling with single-cell resolution in whole animals. Mime-seq overcomes current challenges in cell-type-specific small RNA profiling and provides novel entry points for understanding the function of miRNAs in spatially restricted physiological settings.

Summary:

Introduction: Post-transcriptional silencing by small regulatory RNAs such as miRNAs play essential roles in shaping gene expression during these processes. These 21-24-nt long RNA act in an RNA-induced silence complex (RISC) with an Argonaute-family protein, directing repression of target mRNAs through base-pairing interactions typically between the mRNA 3UTR and the miRNA 5 end or seed region. The functional contribution of miRNA, or other gene expression regulators, to the biology of an organism depends on the cells in which they act. Thus, miRNA profiling methods are powerful entry points to dissect the roles of members of this repressor class.

Results: An Arabidopsis RNA methyltransferase HEN1 methylates animal miRNAs. RNA-methylation is a common modification of small RNAs in plants and animals. This modification is absent from the bulk of animal miRNs. We show that methylation of RNA is regulated by the RNA polymerase Ath-HEN1 in DrosophilaS2 cells. This methylated RNA preserves an adapter-ligatable 3OH for cDNA library preparation.

Discussion: Mime-seq overcomes current challenges forin vivomiRNA profiling from specific cell types within complex tissues or whole organisms. It circumvents excessive manipulations such as cell sorting or immunoprecipitation, which reduce yields and increase noise, achieving greater sensitivity. In direct comparison with neuronal and intestinal Argonaute immunoprespiitations in C. elegans, mime-Seq identified 3- and 2.6-fold more miRNAs respectively, many of which we validated, including a miRNA expressed in a single neuron out of the whole animal.

Figures:

Figure 1: The methyltransferase HEN1 from Arabidopsis thaliana methylates animal miRNAs in vitro and in vivo .

Figure 2: Periodate-mediated oxidation enables selective cloning of animal miRNAs upon restrictive Ath-HEN1-directed methylation.

Figure 3: Mime-seq reproducibly recovers neuronal miRNAs in C. elegans .

Figure 4: Mime-seq unveils the miRNomes of diverse tissues in C. elegans.

Figure 5: Mime-seq has the sensitivity to retrieve miRNAs from 1/300 cells.

Mentioned biomedical entities:

Genes: ASE, Actin, Argonaute, Cas9, Gal4, HEN1, HENN-1, Hen1, RDE-1, Tubulin, Unc, attB, dme, elt-2, hen1, lsy-6, miR-252, miR-277, myo-2, rab-3, rgef-1, tRNAGly

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, neuron

Organisms: Mus musculus, flies, mouse

Tissues: body wall muscle, cuticle, seed, tissue, tube

Pathways: N/A

Query Result 19/25

Developmentally programmed histone H3 expression regulates cellular plasticity at the parental-to-early embryo transition

Gleason, Ryan J.; Guo, Yanrui; Semancik, Christopher S.; Ow, Cindy; Lakshminarayanan, Gitanjali; Chen, Xin

Science Advances

Year: 2023 PMID: 37027463 Citations: 5 Score: 7.05 DOI: 10.1126/sciadv.adh0411

Keywords Hits: lsy-6: 0, che-1: 20

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.003

Abstract:

During metazoan development, the marked change in developmental potential from the parental germline to the embryo raises an important question regarding how the next life cycle is reset. As the basic unit of chromatin, histones are essential for regulating chromatin structure and function and, accordingly, transcription. However, the genome-wide dynamics of the canonical, replication-coupled (RC) histones during gametogenesis and embryogenesis remain unknown. In this study, we use CRISPR-Cas9mediated gene editing in Caenorhabditis elegans to investigate the expression pattern and role of individual RC histone H3 genes and compare them to the histone variant, H3.3. We report a tightly regulated epigenome landscape change from the germline to embryos that are regulated through differential expression of distinct histone gene clusters. Together, this study reveals that a change from a H3.3- to H3-enriched epigenome during embryogenesis restricts developmental plasticity and uncovers distinct roles for individual H3 genes in regulating germline chromatin.

Summary:

INTRODUCTION: Histones, the main protein component of chromatin, are essential to the establishment and maintenance of particular chromaton structures, associated with distinct cell fates. The canonical or replication-coupled histones (i.e., H3, H4, H2A, and H2B) are expressed during S phase and mainly incorporated into the genome during DNA replication. In metazoans, several histone H3 variants have been identified, among which include the replicative and replacement variants, H3 and H3.3, respectively. Histone variants are typically replication-independent (RI), expressed throughout the cell cycle, and regulate a variety of biological processes.

RESULTS: class II histone H3gene clusters (his-40is unique, considering that it can only be detected in a subset of somatic lineages) show distinct transcription patterns during gametogenesis and early embryogenesis. These findings suggest that the histoneH3genome is a complex system with multiple histones. The histone gene clusters are characterized by distinct expression patterns during embryogenesis and embryogenesis, suggesting that the RC histone genes are a major contributor to the epigenetic inheritance of the C. elegans genome.

DISCUSSION: Here, our results demonstrate that the expression of each histone cluster is developmentally regulated for tightly coordinated expression, which is important for embryonic development and germline fitness. Results from other metazoan embryos, such as sea urchin, suggest a potentially conserved regulation of distinct histone gene clusters during embryogenesis. For example, sea urschins have two sets of RC histone genes. From the oocyte stage until gastrulation, histone H3 is expressed from a subset of histone ...

Figures:

Figure 1: Differential incorporation of the RC histone H3 gene family.

Figure 2: H3 and H3.3 occupy distinct chromatin domains in different staged germ cells.

Figure 3: The early embryonic epigenomic landscape is developmentally programmed to switch from histone variant H3.3-enriched to histone H3-enriched upon gastrulation.

Figure 4: Expression patterns of all histone H3 gene clusters and H3.3 genes during early embryonic development.

Figure 5: A dominant-negative mutation in the somatic-specific histone H3 gene, his-6(H113D) , destabilizes the histone tetramer (H3-H4)2, leading to reduced H3 incorporation and extended embryonic plasticity.

Figure 6: The dominant-negative allele of H3, his-6(H113D) , does not significantly change the total nucleosome levels but does lead to reduced levels of H3K27me2/3.

Mentioned biomedical entities:

Genes: ASER, Asp, B, Cas9, H2A, H2B, H3, H3.3, H4, HIRA, HIS3, HIS4, HIS5, Histone, MA, MET-2, P40, PAM, ST65, V5, gcy-5, his-6, histone, hsp16, mes-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, epithelial cell, germ cell, oocyte, somatic cell, sperm, stem cell

Organisms: E. coli, Escherichia coli OP50, Human, Mouse, Xenopus laevis, human, mice, mouse, rat, zebrafish

Tissues: egg, gonad, seed, tissue

Pathways: N/A

Query Result 20/25

The inner junction protein CFAP20 functions in motile and non-motile cilia and is critical for vision

Chrystal, Paul W.; Lambacher, Nils J.; Doucette, Lance P.; Bellingham, James; Schiff, Elena R.; Noel, Nicole C. L.; Li, Chunmei; Tsiropoulou, Sofia; Casey, Geoffrey A.; Zhai, Yi; Nadolski, Nathan J.; Majumder, Mohammed H.; Tagoe, Julia; DEsposito, Fabiana; Cordeiro, Maria Francesca; Downes, Susan; Clayton-Smith, Jill; Ellingford, Jamie; Ambrose, J. C.; Ambrose, J. C.; Arumugam, P.; Bevers, R.; Bleda, M.; Boardman-Pretty, F.; Boustred, C. R.; Brittain, H.; Brown, M. A.; Caulfield, M. J.; Chan, G. C.; Giess, A.; Griffin, J. N.; Hamblin, A.; Henderson, S.; Hubbard, T. J. P.; Jackson, R.; Jones, L. J.; Kasperaviciute, D.; Kayikci, M.; Kousathanas, A.; Lahnstein, L.; Lakey, A.; Leigh, S. E. A.; Leong, I. U. S.; Lopez, F. J.; Maleady-Crowe, F.; McEntagart, M.; Minneci, F.; Mitchell, J.; Moutsianas, L.; Mueller, M.; Murugaesu, N.; Need, A. C.; ODonovan, P.; Odhams, C. A.; Patch, C.; Perez-Gil, D.; Pereira, M. B.; Pullinger, J.; Rahim, T.; Rendon, A.; Rogers, T.; Savage, K.; Sawant, K.; Scott, R. H.; Siddiq, A.; Sieghart, A.; Smith, S. C.; Sosinsky, A.; Stuckey, A.; Tanguy, M.; Taylor Tavares, A. L.; Thomas, E. R. A.; Thompson, S. R.; Tucci, A.; Welland, M. J.; Williams, E.; Witkowska, K.; Wood, S. M.; Zarowiecki, M.; Mahroo, Omar A.; Hocking, Jennifer C.; Cheetham, Michael E.; Webster, Andrew R.; Jansen, Gert; Blacque, Oliver E.; Allison, W. Ted; Au, Ping Yee Billie; MacDonald, Ian M.; Arno, Gavin; Leroux, Michel R.

Nature Communications

Year: 2022 PMID: 36329026 Citations: 16 Score: 6.86 DOI: 10.1038/s41467-022-33820-w

Keywords Hits: lsy-6: 0, che-1: 21

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.002

Abstract:

Motile and non-motile cilia are associated with mutually-exclusive genetic disorders. Motile cilia propel sperm or extracellular fluids, and their dysfunction causes primary ciliary dyskinesia. Non-motile cilia serve as sensory/signalling antennae on most cell types, and their disruption causes single-organ ciliopathies such as retinopathies or multi-system syndromes. CFAP20 is a ciliopathy candidate known to modulate motile cilia in unicellular eukaryotes. We demonstrate that in zebrafish, cfap20 is required for motile cilia function, and in C. elegans, CFAP-20maintains the structural integrity of non-motile cilia inner junctions, influencing sensory-dependent signalling and development. Human patients and zebrafish with CFAP20 mutations both exhibit retinal dystrophy. Hence, CFAP20 functions within a structural/functional hub centered on the inner junction that is shared between motile and non-motile cilia, and is distinct from other ciliopathy-associated domains or macromolecular complexes. Our findings suggest an uncharacterised pathomechanism for retinal dystrophy, and potentially for motile and non-motile ciliopathies in general.

Summary:

Introduction: Motile cilia are structurally conserved across metazoans, being powered by inner and outer dynein arms, whose coordinated movements are regulated by radial spoke and dmynein regulatory complex modules. Motile Cilia are structuraly conserved throughout metazoan life. Mottle Cilia enable sperm locomotion, clear respiratory airways, move cerebrospinal fluid, and contribute to left-right body axis patterning.

Results: cfap20 orthologues have remarkable sequence conservation throughout eukaryotes and are required for motile cilia functions in Paramecium, the single-celled algaeChlamydomonas, andDrosophilasperm Supplementary Fig Yanagisawa et al. also demonstrated that transient knockdown ofcffa20disrupts zebrafish left-right organiser cilii, causes cardiac situs defects, and produces body-axis curvature defects. To build upon this observation, and test if cFap 20 is required formotile cilia functions in a post-larval vertebrate, we mutatedcfp20in zeBRAfish. CRISPR/Cas9 was used to produce a frameshift mutation in exon 3 that predicts a protein disruption at residue 70 (NM200811.1:c.209delGinsTCGAGCTA; p.(Gly70Valfs29 thereby lacking the majority of the highly conserved CFAP20 residues (Fig., denoted allelecfapi20ua5025) that disrupts the single copy and single splice isoform ofcFapi20identifiable in zeBrafish. These data are consistent withcfip20uua525being a null allele or strong hypomorph, and we hereafter refer to the mutants ascfabi20/.

Discussion: CFAP20 is a conserved protein with well-established roles in motile and non-motile metazoan cilia. Our findings support a model whereby CFAPP20 plays a critical role within a structural/functional hub centered in and around the inner junction of ciliary double microtubules (Inner Junction Hub). This region likely regulates both motile cilian beating and non motile (primary) cilial function.

Reporting summary: The following information is available in thelinked to this article: . Further information on research design is availablein thelinkedto this article. . Supplementary information: - . -. .- .-.-.. / --.-- , : () /-, -,,-(-)-/,

Figures:

Figure 1: The two principal classes of ciliopathies are considered to be mutually exclusive: they affect the functions of motile or non-motile (primary) cilia, and they have little to no genetic overlap.

Figure 2: Zebrafish cfap20 mutants display developmental phenotypes characteristic of a motile ciliopathy.

Figure 3: C. elegans CFAP-20 localises to non-motile cilia and is required for the structural integrity of the axoneme inner junction.

Figure 4: C. elegans CFAP20 and PACRG perform non-redundant roles in cilium-dependent behaviours and development: gustatory plasticity, lifespan control, locomotion, and body size.

Figure 5: CFAP20 variants segregate with autosomal recessive retinal dystrophy.

Figure 6: CFAP20 patient variant sequences variously cause altered protein stability, mislocalisation in C. elegans and reduced ability to rescue zebrafish homozygote development.

Figure 7: Zebrafish cfap20 / mutants model early-onset, progressive retinal dystrophy.

Figure 8: Model depicting the Inner Junction Hub (IJH) as regulatory domain in motile cilia and previously uncharacterised ciliopathy (retinal dystrophy) region that influences the signalling functions of non-motile (primary) cilia.

Mentioned biomedical entities:

Genes: ARL3, ARL6, ASE, BBS1, BBS2, C, CA, CEP290, CFAP52, CHE-11, CNGA1, EFHC1, ER, ERG, IFT140, Lab, MA, MO, Myc, NE, OSM-3, PACRG, PCRG-1, RPGR, SEM, SP6, Salt, Sim, TEKT1, TULP1, WT, WTs, XBX-1, foxa3, myl7, pcrg-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, sperm

Organisms: C. elegans, Human, Zebrafish, fish, human, humans, mouse, zebrafish

Tissues: Photoreceptor, Tissue, cone, cornea, interneuron, macula, photoreceptor, retinal pigment epithelium, sperm, synapse, tissue, yolk

Pathways: N/A

Query Result 21/25

Guidelines for monitoring autophagy in Caenorhabditis elegans

Zhang, Hong; Chang, Jessica T; Guo, Bin; Hansen, Malene; Jia, Kailiang; Kovcs, Attila L; Kumsta, Caroline; Lapierre, Louis R; Legouis, Renaud; Lin, Long; Lu, Qun; Melndez, Alicia; ORourke, Eyleen J; Sato, Ken; Sato, Miyuki; Wang, Xiaochen; Wu, Fan

Autophagy

Year: 2015 PMID: 25569839 Citations: 98 Score: 6.67 DOI: 10.1080/15548627.2014.1003478

Keywords Hits: lsy-6: 5, che-1: 0

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.000

Abstract:

The cellular recycling process of autophagy has been extensively characterized with standard assays in yeast and mammalian cell lines. In multicellular organisms, numerous external and internal factors differentially affect autophagy activity in specific cell types throughout the stages of organismal ontogeny, adding complexity to the analysis of autophagy in these metazoans. Here we summarize currently available assays for monitoring the autophagic process in the nematode C. elegans. A combination of measuring levels of the lipidated Atg8 ortholog LGG-1, degradation of well-characterized autophagic substrates such as germline P granule components and the SQSTM1/p62 ortholog SQST-1, expression of autophagic genes and electron microscopy analysis of autophagic structures are presently the most informative, yet steady-state, approaches available to assess autophagy levels in C. elegans. We also review how altered autophagy activity affects a variety of biological processes in C. elegans such as L1 survival under starvation conditions, dauer formation, aging, and cell death, as well as neuronal cell specification. Taken together, C. elegans is emerging as a powerful model organism to monitor autophagy while evaluating important physiological roles for autophagy in key developmental events as well as during adulthood.

Summary:

Conclusion and Perspectives: Recent studies have established C. elegans as a powerful genetic model to delineate the autophagy pathway, such as through the identification of metazoan-specific autophagic genes. Moreover, C.elegansoncologyhas emerged as an excellent model system to reveal the functions of autophagia under physiological and stress conditions at the subcellular, tissue-specific, and organismal levels. For these studies, multiple assays for monitoring the autophagy process have been developed, yet it is important to note that each assay, summarized above, has its limitations and thus a combination of these assay(s) should always be performed to make solid claims about autophagical activity.

Figures:

Figure 1: The hierarchical order of autophagy genes in the aggrephagy pathway.

Figure 2: LGG-1 pattern during embryogenesis. (A) LGG-1 precursor, LGG-1-I (unlipidated form) and LGG-1-II (lipidated form) in wild-type and atg-4.1, epg-4 and atg-3 mutant embryos. (BE) LGG-1 forms punctate structures at the 100-cell stage ((B) and C), but these structures disappear at the comma stage in wild-type embryos ((D) and E). ((B) and D) DAPI images of the embryos shown in ((C) and E), respectively. ((F) and G) LGG-1 puncta are absent in atg-3 embryos. (F) DAPI image of the embryo shown in (G). (H) LGG-1 accumulates into enlarged cluster-like structures in epg-4 embryos. (IL) GFP::LGG-1 is largely diffuse in the cytoplasm with some punctate structures. ((I) and K) DIC images of the embryos shown in ((J) and L), respectively. (MP) GFP::LGG-1 forms aggregates in atg-3 mutant embryos, which colocalize with SQST-1 aggregates. Inserts show magnified views. (M) DAPI image of the embryo shown in (NP). Scale bar: 5m (B-P); 2.5m (inserts in NP). C. elegans embryos remain the same size during embryogenesis. Thus, the scale bar is only shown once in each figure.

Figure 3: LGG-2 puncta during embryogenesis. ((A) and B) Confocal images of LGG-1 (red) and LGG-2 (green) in wild-type and epg-3 1-cell embryos. (C) Western blot of protein extracts from mixed-stage GFP::LGG-2 and GFP::LGG-2 G130A transgenic worms incubated with anti-GFP antibody. (D) Electron micrographs of GFP::LGG-2 embryos incubated with anti-GFP antibodies. The right panel shows a magnified view of the autophagosome. Scale bar: 200nm. ((E) and F) Confocal images of GFP::LGG-2 and GFP::LGG-2 G130A in 500-cell embryos. GFP::LGG-2 localizes in both a punctate and a diffuse pattern while GFP::LGG-2 G130A is diffusely localized in the cytosol. This figure was previously published in references 24 and 25 and is reproduced by permission of Elsevier and Landes Bioscience.

Figure 4: Various protein aggregates are degraded by autophagy during embryogenesis. ((A) and B) PGL-1 granules are restricted to germ precursor cells Z2 and Z3 in wild-type embryos. ((C) and D) PGL-1 granules accumulate in somatic cells in atg-3 mutant embryos. ((E) and F) SEPA-1 forms spherical aggregates at the 100 cell stage in wild-type embryos. ((G) and H) SEPA-1 aggregates disappear in wild-type comma stage embryos. ((I) and J) SEPA-1 aggregates accumulate in atg-3 mutant embryos throughout embryogenesis. ((K) and L) No SQST-1 aggregates are detected in wild-type embryos. ((M) and N) SQST-1 accumulates into numerous aggregates in atg-3 mutant embryos. (A, C, E, G, I, K, and M) DAPI images of the animals shown in (B, D, F, H, J, L, and N), respectively. Scale bar: 5m (A-N).

Figure 5: GFP::LGG-1 pattern at post-embryonic stages. (A and B) GFP::LGG-1 is diffusely localized in the hypodermis of wild-type animals. (A) DIC image of the animal shown in (B). (C) GFP::LGG-1 forms numerous small punctate structures in the hypodermis after the animals are starved

for 4h. (D and E) GFP::LGG-1 is diffusely localized in seam cells of wild-type animals. (D) DIC image of the animal shown in (E). (F) GFP::LGG-1 forms many small punctate structures in the seam cells after the animals are starved for 4h. Inserts show magnified views. (G) Numerous GFP::LGG-1 punctate structures are observed in the intestine of epg-5 mutants. L4 larvae are shown in (A-G). (H and I) A large number of GFP::LGG-1 puncta accumulate in atg-3 mutants at the L1 larval stage (H), but largely disappear at late larval stage (I). Scale bars: 20m (AC, G-I); 10m (DF); 10m (inserts in (B and C); 5m (inserts in (E and F).

Figure 6: Bafilomycin A 1 treatment increases GFP::LGG-1 puncta in seam cells of adult animals. (A) A few GFP::LGG-1 puncta form in seam cells in DMSO-treated wild-type day 1 adult animals. (B) The number of GFP::LGG-1 puncta dramatically increases after bafilomycin A 1 (Baf A) treatment for 2h in wild-type day 1 adult animals. Arrows point to puncta. Scale bar: 5m.

Figure 7: GFP::LGG-1 pattern in L3 larva using different mounting methods. (A) GFP::LGG-1 is largely diffuse in seam cells of L3 larvae mounted with M9 medium on a 2% agarose pad. These animals are not paralyzed. (B) GFP::LGG-1 is largely diffuse in seam cells of fully anesthetized L3 larvae mounted in a final concentration of 0.1 % (w/v; 150mM) NaN 3 in M9 medium on a 2% agarose pad with the same concentration of NaN 3 . (CD) GFP::LGG-1 forms many different-sized punctate structures (arrows) in the seam cells (and other tissues) of fully anesthetized L3 larvae mounted in a final concentration of 0.2mM (C), or 2mM (D) tetramisole in M9 medium on a 2% agarose pad. Punctate structures start appearing 5min after mounting. (EF) GFP::LGG-1 forms many different-sized punctate structures (arrows) in the seam cells (and other tissues) of fully anesthetized LC3 larvae mounted in a final concentration of 0.2mM (C) or 2mM (D) levamisole in M9 medium on a 2% agarose pad. Punctate structures start appearing 5min after mounting. Scale bar: 5m.

Figure 8: Expression of SQST-1 aggregates at larval stages. (A and B) SQST-1::GFP is weakly expressed in wild-type L4 larvae. bpIs151(sqst-1p:: sqst-1::gfp) was used. (C and D) In epg-9 mutants, SQST-1::GFP accumulates into aggregates in multiple tissues. (A and C) DAPI images of the animals shown in (B and D), respectively. Scale bar: 20m.

Figure 9: Accumulation of SQST-1 aggregates in the intestine in rpl-43 mutants. (A and B) SQST-1::GFP accumulates into a large number of aggregates in the intestine in rpl-43 mutant L4 larvae. (C-F) Accumulation of SQST-1 aggregates in rpl-43 mutant L4 larvae is suppressed by elevated autophagy activity induced by starvation (C and D) or by lin-35 inactivation (E and F). (A, C, and E) DIC images of the animals shown in (B, D, and F), respectively. Scale bar: 20m (AF).

Figure 10: HLH-30::GFP translocates to the nucleus upon LET-363/TOR inactivation in adult animals. (A) HLH-30::GFP shows diffuse localization in the cytoplasm in the intestine of day 1 animals. (B) HLH-30::GFP translocates to the nucleus following whole-life let-363/TOR RNAi treatment. Scale bar: 300m.

Figure 11: Electron microscopy images of autophagic structures in wild-type and autophagy mutants. (AD) The 3 main autophagic structures (subcompartments): (A) phagophore; (B) autophagosome; (C) light-type, actively digesting autolysosome; (D) autolysosomes with dark undigested contentthey may frequently have irregular shapes. (E) Complex myelinated autophagic structure in a hypodermal cell of an unc-51(e361) mutant. (FH) Three types of secretory vacuoles, which are similar to autolysosomes. Scale bars: 500nm.

Mentioned biomedical entities:

Genes: ARC, ASE, ASER, ATG-4.2, ATG-7, ATG16, ATG8, Atg1, Atg10, Atg12, Atg13, Atg16, Atg4, Atg5, Atg7, Atg8, CEA, ER, F44F1.6, FL, GABARAP, HLH-30, INS, L1, LC3, LGG-1, LGG-2, MO, MOs, P3, P4, PE, PGL-1, PGL-3, RI, SEPA-1, SQST-1, SQSTM1, TFEB, Vps34, WNT, act-1, ama-1, atg-16.2, atg-18, atg-5, atg-7, bec-1, cog-1, cyn-1, daf-2, hgrs-1, lgg-1, lgg-2, lsy-6, nhr-23, pmp-3, unc-51

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, sperm

Organisms: C. elegans, human

Tissues: cuticle, eggshell, epidermis, hypodermis, neuron, oviduct, spermatheca, yolk

Pathways: N/A

Query Result 22/25

Neurogenesis in Caenorhabditis elegans

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Genetics

Year: 2024 PMID: 39167071 Citations: 11 Score: 6.30 DOI: 10.1093/genetics/iyae116

Keywords Hits: lsy-6: 17, che-1: 34

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.001

Abstract:

Animals rely on their nervous systems to process sensory inputs, integrate these with internal signals, and produce behavioral outputs. This is enabled by the highly specialized morphologies and functions of neurons. Neuronal cells share multiple structural and physiological features, but they also come in a large diversity of types or classes that give the nervous system its broad range of functions and plasticity. This diversity, first recognized over a century ago, spurred classification efforts based on morphology, function, and molecular criteria. Caenorhabditis elegans, with its precisely mapped nervous system at the anatomical level, an extensive molecular description of most of its neurons, and its genetic amenability, has been a prime model for understanding how neurons develop and diversify at a mechanistic level. Here, we review the gene regulatory mechanisms driving neurogenesis and the diversification of neuron classes and subclasses in C. elegans. We discuss our current understanding of the specification of neuronal progenitors and their differentiation in terms of the transcription factors involved and ensuing changes in gene expression and chromatin landscape. The central theme that has emerged is that the identity of a neuron is defined by modules of gene batteries that are under control of parallel yet interconnected regulatory mechanisms. We focus on how, to achieve these terminal identities, cells integrate information along their developmental lineages. Moreover, we discuss how neurons are diversified postembryonically in a time-, genetic sex-, and activity-dependent manner. Finally, we discuss how the understanding of neuronal development can provide insights into the evolution of neuronal diversity.

Summary:

Introduction: The complex functionalities of animal nervous systems rely on a vast diversity of neurons, which receive sensory inputs, integrate different internal and external signals, and produce cognitive and behavioral outputs. Neurons are highly specialized cells, with unique morphological and physiological properties and an outstanding capacity for intercellular communication. While they share several common functional and molecular properties, they present in a multitude of different types or classes. Cellular diversity within nervous systems was recognized more than 100 years ago when the remarkable range of differences in neuronal morphologies was revealed and prompted multiple efforts toward neuronical classification, initially based on morphology but later including functional and m

Concluding remarks: The last 2 decades of studies on neuron specification and differentiation of the complete nervous system ofC. eleganshave allowed us to start building a holistic model of the regulatory networks involved in neurogenesis, with a degree of molecular and cellular resolution that is difficult to attain in other animal models. We have distilled the vast genetic and molecule analyses generated by theC.elegansoncology, described here, into the regulatory framework summarized in. This view also highlights open questions.

Figures:

Figure 1: Regulatory framework for C. elegans neurogenesis. Summary of the main regulatory interactions controlling neurogenesis and neuron diversification that should serve as a roadmap for this chapter. Development of each neuron class requires integration of regulatory information from different developmental timepoints into parallel, yet interconnected regulatory modules that coexist in the cell. (1) Lineage history determines the set of TFs expressed in the neuron and its progenitors, the signalling events it receives over time, and the particular chromatin landscape of the postmitotic neuron (see Lineage-based mechanisms of neuronal diversification). (2) bHLH proneural factors have evolutionary conserved roles in the specification of neuronal progenitors (see Diverse functions of bHLH proneural TFs). (3) Terminal selector TFs act in the postmitotic neuron to directly activate broad neuron class effector gene batteries (see Rich sets of effector genes distinguish different neuron classes and Principles of neuron class specification by terminal selectors). (4) Terminal selectors act together with HOX TFs to diversify some neuron subclasses along the AP axis (see Neuronal diversification across body axes). (5) Neuron class transcriptomes are also shaped by postembryonic time or genetic sex (see Neuronal diversification over developmental time and across sexes). (6) Mature neurons can modify their transcriptome in response to environmental stimuli (see Environmental effects on neuron gene expression). (7 and 8) At least 2 regulatory modules run in parallel to those specifying neuron class properties: panneuronal effector genes, those shared by all neuron classes are under direct control of CUT HD TFs (7) (see Panneuronal features define neurons as a tissue type) while different TFs activate the ciliome components expressed by all sensory ciliated neurons (8) (see Rich sets of effector genes distinguish different neuron classes). t.g. , neuron type effector gene ; p.g. , panneuronal effector gene ; c.g. , ciliome effector gene .

Figure 2: Developmental origin of neurons and neuronal diversity. a) Matrix of the 118 morphologically different neuron classes in hermaphrodite C. elegans classified according to different criteria including embryonic/postembryonic lineage, function, and neurotransmitter identity. For simplicity, large effector gene categories, such as neuropeptides or innexins, have not been included but references to their expression patterns are provided in

the text. b) Lineage origins of all 302 embryonically and postembryonically generated hermaphrodite neurons. c) A cartoon illustration of a generic neuron depicting different sets of proteins required for specific neuronal functions, which include panneuronal factors such as synaptic components, cilium proteins (ciliome) present in ciliated sensory neurons, and different sets of neuron class-specific effector proteins such as adhesion molecules, enzymes, ion channels, receptors, or neuropeptides.

Figure 3: Progressive determination model of neuronal development. A cartoon illustration of the progressive determination model of neural development. Key steps relevant to C. elegans include the specification of neural progenitors and the differentiation of postmitotic cells into neurons.

Figure 4: Expression of bHLH TFs in neuronal lineages. a) Summary of proneural gene expression at any point in the lineage of the indicated embryonic neurons (sources: WormAtlas, Sulston and Horvitz 1977 ; Sulston et al. 1983 ; Reilly et al. 2022 ; Masoudi et al. 2023). b) An illustrative example of sequential bHLH expression in the ABalpp/ABpraa lineage. hlh-3 is expressed both early and broadly and has no known role in this lineage. hlh-14 is expressed in a more restricted manner and is required for neural precursor specification. hlh-4 is only expressed in the ADL neurons where it acts as a terminal selector and not as proneural gene. See the main text for more details.

Figure 5: Terminal selector functions in neuronal diversification. a) Example of the accumulation of regulatory factors along a lineage to result in specific (and robust) patterns of gene expression (from Walton et al. 2015). b) Terminal selector TFs act combinatorially on modular cis- regulatory sequences, enabling cell-specific control of various terminal effector genes. The modular architecture of cis- regulatory elements enables integration of distinct combinations of terminal selectors, as well as integration with other transcriptional inputs as discussed in the rest of the chapter. The 3 illustrated examples correspond to those described in the text. c) The output of a terminal selector (e.g. UNC-86) can be diversified by integration into gene regulatory networks with other transcriptional regulators. These can be organized in regulatory motifs such as coherent feedforward loops (in which a terminal selector activates an effector gene as well as an additional transcriptional activator, blue arrows) and incoherent feedforward loops (in which a terminal selector activates an effector gene as well as a repressor of the same target gene, red arrows). Such motifs are recurrently used to modify the terminal battery activated by a terminal selector. d) Regulation of the different components of the neuronal transcriptome is achieved thanks to combinatorial TF action on modular cis- regulatory sequences that result in parallel modules that are interconnected. For example, panneuronal gene expression relies on CUT TFs but also receives terminal selector input. Gene modules such as those coding for the structural cilium components are under control of a parallel program driven by FKH-8 and DAF-19C, but other neuron class-specific components of the cilia, just like other neuron-type effector genes, are under terminal selector control.

Figure 6: Generation of terminal selector combinations in space and time. a) Schematic of how the Wnt asymmetry pathway can be integrated with existing TFs to generate a novel regulatory state during every cell division (adapted from Bertrand 2016). The asymmetry in POP-1 and SYS-1 results in distinct transcriptional outputs in the anterior vs the posterior daughter cell. Note that this schematic is highly simplified and POP-1 in the anterior cell can also activate or repress targets (Murgan and Bertrand 2015). b) Specification of the AIY neurons illustrates a number of points described in the text. The vertical axis denotes time. Transient expression of a trigger, REF-2, is integrated with the Wnt asymmetry pathway and autoactivation of the terminal selectors TTX-3 and CEH-10 to result in specific and robust activation in the posterior cell. This distinguishes the SMDD (anterior) from the AIY (posterior) neurons. c) The autoactivation mechanism of che-1 has been proposed to be sensitive to fluctuations in the CHE-1 protein by preferential activation of its own locus (Traets et al. 2021). d) The differentiation of the left and right ASE neurons also illustrates a number of mechanisms described in the text. These 2 neurons derive from lineages that diverge at the 4-cell stage owing to the first Notch induction and represent one of the best described examples of convergence. The point where the lineages converge is marked with a dashed line. The converging branches generate 11 pairs of bilaterally symmetric neurons and 2 pairs of bilaterally symmetric glia. The early lineage asymmetry is integrated with the terminal selector CHE-1 in the form of differential chromatin states of the lsy-6 locus. This determines whether lsy-6 is expressed (ASEL) or not (ASER) and the downstream acquisition of the respective terminal gene batteries (Hobert 2014) (Fig. 7a).

Figure 7: Repressor-based mechanisms for neuronal diversification. a) Schematic of the double-repressor interactions that result in ASE asymmetry. The miRNA lsy-6 in ASEL represses the production of the TF COG-1. cog-1 is transcribed in both ASEs but only produces COG-1 protein in ASER where lsy-6 is absent. COG-1 acts with the corepressor Groucho/UNC-37 to repress transcription of die-1 , which encodes another TF. The mutually exclusive expression of COG-1 and DIE-1 define downstream molecular asymmetries that determine the asymmetric sensory function of the 2 ASEs. b) Scheme of the repressor-based diversification of distinct motor neuron classes. All classes rely on the terminal selector UNC-3 for their differentiation, but each class expresses a distinct combination of repressors that counteract the action of UNC-3 on specific gene batteries (Kerk et al. 2017). c) The action of the terminal selector UNC-86 in the ALM and BDU sister neurons is diversified by the asymmetric expression of MEC-3 in ALM. In the absence of MEC-3, UNC-86 acts cooperatively with PAG-3 to activate BDU-specific gene expression. In ALM, MEC-3 binds UNC-86, which now activates the ALM-specific battery and is titrated away from PAG-3-dependent genes (Gordon and Hobert 2015). d) The miRNA miR-791 is specifically expressed in the BAG, AFD, and ASE neurons, where it posttranscriptionally represses 2 otherwise ubiquitously transcribed genes, cah-3 and akap-1 . This repression is necessary in the BAG neurons to elicit the correct avoidance response to high CO 2 levels (Drexel et al. 2016). e) Key components of the heterochronic pathway and their regulatory interactions (Ambros 2000 ; Ketting and Cochella 2021). Proteins are in blue; regulatory RNAs are in red and include the miRNAs lin-4 and let-7 and the lep-5 lncRNA that associates with Makorin/LEP-2 (Herrera et al. 2016 ;

Kiontke et al. 2019). LIN-14, HBL-1, and LIN-29 are TF outputs of the pathway that are integrated with other TFs (e.g. terminal selectors) on the cis -regulatory regions of neuronal genes discussed in the text.

Figure 8: Environmental effects on neuron diversity. a) Different environmental triggers are integrated into gene expression changes through parallel modules acting in the neuron. b) L4 expression data of signal activated TFs and some examples of terminal selectors in sensory neurons. Signal-activated TFs, such as MEF-2, CRH-1, and DAF-16, are broadly expressed in the nervous system; however, they produce neuron class-specific responses acting together with terminal selectors that show a more restricted pattern of expression. Data source CENGEN. c) The 2 illustrated examples correspond to those described in the text. CRMs of plastic effector genes can integrate input from signal-regulated TFs as well as terminal selectors. ADL expression of srh-234 is directly regulated by the HLH-4 terminal selector but upon starvation is repressed by the MEF-2 TF. tph-1 expression in the ADF neuron is plastic to several environmental conditions, and pathogen exposure induces tph-1 expression through the activity of the LAG-1 ADF terminal selector.

Mentioned biomedical entities:

Genes: ADF, ALN, ASE, ASER, AST-1, B, BAG, BNC-1, CEH-10, CEH-13, CEH-14, CEH-20, CEH-34, CEH-36, CHE-1, COG-1, CRH-1, CUT, CtBP, DA, DAF-16, DAF-19, DAs, DB, DDs, DIE-1, DMD-3, DR, Dop, EGL-44, EGL-5, FKH-8, FLP, Glu, Groucho, HES, HLH-2, Hh, I3, LAG-1, LIN-11, LIN-14, LIN-26, LIN-28, LIN-29, LIN-39, LIN-41, LIN-53, MAB-3, MEC-3, MEF-2, MyT1, MyoD, NGF, NHR, Neurogenin, Notch, ONECUT, Olig, P38, PA, PAG-3, PHB, PHP-3, POP-1, REF-1, REF-2, RIM, RIP, SAB, SOX-2, STR-2, SYS-1, Sox, Sox-3, Sox2, SoxB, SoxC, TBX-37, TF, TRA-1, TTX-3, UNC-130, UNC-18, UNC-3, UNC-30, UNC-39, UNC-4, UNC-55, UNC-86, V5, VAB-7, VCs, VD, VNC, Wnt, ZAG-1, bas-1, cah-3, ced-3, ceh-13, ceh-30, ceh-36, ceh-6, ces-1, ces-2, cog-1, die-1, egl-44, egl-46, egl-5, ham-1, histone, hlh-16, hlh-17, hlh-3, hlh-6, let-7, lin-14, lin-26, lin-28, lin-32, lin-39, lin-4, lsy-6, mab-5, ngn-1, nob-1, odr-10, php-3, sox-2, str-2, unc-3, unc-30, unc-39, unc-4, unc-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, glial cell, interneuron, muscle cell, neural progenitor cell, neural stem cell, neuron

Organisms: C. elegans, cnidarians, humans, mice, mouse, zebrafish

Tissues: blastomere, body wall muscle, egg, hypodermis, neural tissue, neuron, synapse, tissue, ventral

Pathways: N/A

Query Result 23/25

TEG-1 CD2BP2 controls miRNA levels by regulating miRISC stability in C. elegans and human cells

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Nucleic Acids Research

Year: 2017 PMID: 28180320 Citations: 8 Score: 5.02 DOI: 10.1093/nar/gkw836

Keywords Hits: lsy-6: 14, che-1: 0

Keywords Frequency (TF): lsy-6: 0.002; che-1: 0.000

Abstract:

MiRNAs post-transcriptionally regulate gene expression by recruiting the miRNA-induced silencing complex (miRISC) to target mRNAs. However, the mechanisms by which miRISC components are maintained at appropriate levels for proper function are largely unknown. Here, we demonstrate that Caenorhabditis elegans TEG-1 regulates the stability of two miRISC effectors, VIG-1 and ALG-1, which in turn affects the abundance of miRNAs in various families. We demonstrate that TEG-1 physically interacts with VIG-1, and complexes with mature let-7 miRNA. Also, loss of teg-1 in vivo phenocopies heterochronic defects observed in let-7 mutants, suggesting the association of TEG-1 with miRISC is necessary for let-7 to function properly during development. Loss of TEG-1 function also affects the abundance and function of other microRNAs, suggesting that TEG-1s role is not specific to let-7. We further demonstrate that the human orthologs of TEG-1, VIG-1 and ALG-1 (CD2BP2, SERBP1/PAI-RBP1 and AGO2) are found in a complex in HeLa cells, and knockdown of CD2BP2 results in reduced miRNA levels; therefore, TEG-1s role in affecting miRNA levels and function is likely conserved. Together, these data demonstrate that TEG-1 CD2BP2 stabilizes miRISC and mature miRNAs, maintaining them at levels necessary to properly regulate target gene expression.

Summary:

INTRODUCTION: The identification of non-coding RNAs (ncRNAs) and the elucidation of their cellular functions have enhanced our understanding of post-transcriptional gene regulation. The microRNA (miRNA) family of ncARNs consists of 2124 nucleotide-long RNA that bind to complimentary sequences in messenger RNA(mRNA) and regulate their expression. MiRNAs were first identified in Caenorhabditis elegans with lin-4 and let-7 being the first miRNAs described to coordinate temporal developmental events ,). Since then, miRNA(s) have been identified in most plants and animals studied), and are predicted to regulate over half of human genes). Additionally, mis-regulation of miRNA has been implicated in many human diseases, including neuro-degeneration, cardiovascular disease and cancer).

RESULTS: teg-1 mutants display heterochronic defects that are not observed in other splicing factor mutants. Similarly, the mammalian ortholog of TEG-1 interacts with other non-spliceosomal proteins, and has non- slicing functions. Thus, it is likely that the somatic phenotypes observed in integ-1mutants are due to non-Splicing functions of TEU-1.

DISCUSSION: In C. elegans, mis-regulation of miRNA levels often results in defective temporal developmental events. Here, we show that loss of teg-1function leads to a range of developmental defects, and the levels of miRNs from diverse miRNA families are lowered in tte-1mutants. In addition, TEG-1 physically interacts with the miRISC protein VIG-1, and TEG-1/ALG-1 complexes with mature let-7 miRNA and the core miRIS C Argonaute protein, ALG-1. Importantly, the protein levels, but not the mRNA levels, of VIG-7 and ALG-1/a are reduced in a t-teG-1 mutant. These data provide the first evidence of TEG as a miRISc interacting protein and a regulator of miRAN function.

Figures:

Figure 1: teg-1 mutants display heterochronic phenotypes. Mutations in teg-1 lead to somatic phenotypes, such as abnormal vulva morphology (A), defective seam cell syncytium formation (B) and over-replication of seam cells (C). (A) 62% of teg-1(oz230) mutants have a mild protruding vulva phenotype (n = 88), whereas 100% of the N2 (wild-type) animals do not (n = 50). Arrow head indicates the location of vulva. (B) Incomplete fusion of lateral seam cells in teg-1 mutants results in gaps in the seam cell syncytium. Scale bar = 50 m. (C) Over-replication of hypodermal seam cells results in formation of paired nuclei marked by arrows in teg-1 mutants. Each control animal has exactly 16 seam cells, whereas an average of 18 and 20 seam cells were found in teg-1(oz189) and teg-1(oz230) mutants, respectively. Seam cell membranes (B) and nuclei (C) are marked by AJM-1::GFP and SCM::GFP, respectively.

Figure 2: TEG-1 function is required for maintaining mature miRNA levels and function. (A) Reduced mature let-7, lin-4, miR-48, miR-58, miR-61 and miR-62 levels are found in teg-1(oz230) mutants by quantitative real time PCR using TaqMan probes. (B) Northern blot detection of let-7 and miR-58 RNAs in various genetic backgrounds of late L4-stage animals. The detection of tRNA Gly was served as a loading control. (C) Ectopic expression of lsy-6 miRNA from the cog-1 promoter suppresses COG-1::GFP expression in the uterus. However, COG-1::GFP expression is partially restored upon removal of TEG-1 in lsy-6 ectopic expression background (vul = vulva and u = uterus). (D) Quantification of uterine COG-1::GFP expression in various genetic backgrounds. In (A) and (D), the error bars indicate standard deviation, and two-tailed Students t -tests were used to determine the statistical significance: P 0.0005.

Figure 3: TEG-1 associates with VIG-1 and miRISC. (A) A subset of TEG-1 is detected in association with VIG-1 by reciprocal co-IP. In the TEG-1 IP,

VIG-1 is detected with VIG-1-specific antibodies, but not with the pre-immune serum. Conversely, in the VIG-1 IP, TEG-1 is co-IPd, whereas TEG-1 is not detected in an IP using pre-immune rabbit serum. (B) A subset of VIG-1 is detected in association with TAP-tagged TEG-1 using TEG-1 IP followed by detection with VIG-1-specific antibodies in both the presence and absence of RNase A. (C and D) Physical interaction of TEG-1 with VIG-1 using bacterial pull-down experiments followed by immunoblotting. 6X-His-VIG-1 was co-expressed in E. coli with GST, GST-TEG-1 (full length), or GST-TEG-1 with the GYF domain removed (GST-TEG-1(GYF)). When a Ni column was used to pull-down the 6X-His-VIG-1 fusion protein, both full-length and GYF-truncated GST-TEG-1, but not GST alone, were co-precipitated (C). Similarly, VIG-1 was pulled-down from extracts containing full-length and GYF-truncated TEG-1 in the presence and absence of RNase A, using GST beads to pull down GST-TEG-1 fusion proteins (D). (E) TEG-1 associates with a complex containing the mature let-7 miRNA. Immunoblotting analysis using TEG-1-specific antibodies detects the presence of TEG-1 only in the let-7 complementary, but not in the unrelated, 2- O -methylated oligonucleotide IPs.

Figure 4: VIG-1 and ALG-1 protein levels are reduced in teg-1 mutants. (A) A representative immunoblot showing lower TEG-1 protein levels in whole worm extracts from vig-1(tm3383) animals, and lower VIG-1 and ALG-1 protein levels in teg-1(oz230) extracts, with paramyosin as a loading control. The reduction of VIG-1 and ALG-1 levels can be restored in teg-1 mutants expressing exogenous TAP-tagged TEG-1 fusion protein. Approximately 100 one day past L4-staged animals were loaded per lane. (B) A roughly three fold reduction of VIG-1 and ALG-1 protein levels are detected in teg-1(oz230) extracts in comparison to the wild-type N2 extracts by intensity measurement using data collected from three independent experiments. Using paramyosin as an internal control, intensities of the paramyosin bands in N2 (wild-type) samples were set to 100%, whereas the intensities of the corresponding experimental samples were normalized to this value. A two-tailed student t-test was used to determine the statistical significance: P 0.002. (C) vig-1 and alg-1 mRNA levels in teg-1(oz230) mutants are similar to wild-type levels using relative quantitative real time PCR. (DG) Reduced VIG-1 protein levels in both germ line (D and E) and somatic (F and G) tissues of teg-1(oz230) mutants. Indirect immunofluoresence staining of VIG-1 was performed on dissected gonads (D and E) and intestine (F and G) of N2 (wild-type) and teg-1(oz230) animals. UAF-2 was used as control. Images were collected with identical exposure times and the intensity was measured by ImageJ software. Fluorescence intensities of UAF-2 in N2 samples were set to 100%, whereas fluorescence intensities of the experimental samples were normalized to this value. Error bars = standard deviation. P 0.001. Scale bar = 25 m.

Figure 5: VIG-1 protein levels are regulated by proteasome ubiquitin degradation pathway. (A) Increased VIG-1 and ALG-1 protein levels are detected in animals treated with lactacystin, a specific inhibitor for the 26S proteasome. (B and C) Intensity measurements of VIG-1 and ALG-1 protein levels were averaged from three independent experiments. Both VIG-1 and ALG-1 protein levels are increased 1.5-fold upon lactacystin treatment in N2 (B) and teg-1(oz230) (C). A two-tailed student t-test was used to determine the statistical significance: P 0.01. Error bars in B and C = standard deviation. (D) VIG-1 is ubiquitylated. VIG-1 was immunoprecipitated from wild-type extracts in the presence or absence of lactacystin followed by immunoblotting with ubiquitin-specific antibodies. Bands consistent in size with ubiquitylated VIG-1 are detected and become more intense with lactacystin treatment. The two bands labeled as Ub-VIG-1 at 60 kDa and 70 kDa (arrow heads) are consistent in size with mono- and di-ubiquitylated VIG-1, respectively; whereas the poly-ubiquitylated VIG-1 is detected in the upper smear region. The lower panel represents the results of re-probing the membrane with anti-VIG-1 antibodies.

Figure 6: Association of TEG-1 with miRISC proteins is conserved in human tissue culture cells. (A) SERBP1/PAI-RBP1 (VIG-1 ortholog) and AGO2 (ALG-1 ortholog) are co-IPd with CD2BP2 (TEG-1 ortholog). GFP or GFP-CD2BP2 was transiently expressed in HeLa cells and proteins were co-IPd using anti-GFP antibodies. SERBP1/PAI-RBP1 and AGO2 are detected when IPd with anti-GFP antibodies followed by immunoblotting, while the negative control, actin, was not detected. (B) Mature let-7a, miR-24, and miR-26a miRNAs are significantly reduced in HeLa cells treated with CD2BP2 siRNA in comparison with a scrambled siRNA treatment by quantitative real time PCR using TaqMan probes. A two-tailed Students t -test was used to determine the statistical significance: P 0.0003. Error bar = standard deviation. (C) Proposed model. Free miRNAs are degraded by exonucleases (XRN-1/XRN-2), while miRISC proteins are subjected to proteasomal degradation. Our model suggests that TEG-1 CD2BP2 interacts with miRISC. TEG-1 CD2BP2 helps stabilize miRISC protein components. In the absence of TEG-1 CD2BP2, miRISC protein components are more susceptible to proteasomal degradation (thick arrow), which in turn results in decreased stability of miRNAs.

Mentioned biomedical entities:

Genes: AGO2, AIN-1, AIN-2, ALG-1, ALG-2, ASER, Ago2, Argonaute, CD2BP2, CGH-1, COG-1, DCS-1, Dicer, GST, GYF, IPs, LIN-41, NHL-2, SERBP1, TEG-1, U48, VIG-1, XRN-1, XRN-2, actin, alg-1, cog-1, let-7, lin-4, lin-41, lsy-6, miR-84, tRNAGly, vig-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, seam cell, stem cell

Organisms: E. coli, Escherichia coli, human, humans

Tissues: hypodermis

Pathways: N/A

Query Result 24/25

Neuron cilia restrain glial KCC-3 to a microdomain to regulate multisensory processing

Ray, Sneha; Gurung, Pralaksha; Manning, R. Sean; Kravchuk, Alexandra A.; Singhvi, Aakanksha

Cell reports

Year: 2024 PMID: 38421867 Citations: 9 Score: 3.32 DOI: 10.1016/j.celrep.2024.113844

Keywords Hits: lsy-6: 0, che-1: 6

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.001

Abstract:

Glia interact with multiple neurons, but it is unclear whether their interactions with each neuron are different. Our interrogation at single-cell resolution reveals that a single glial cell exhibits specificity in its interactions with different contacting neurons. Briefly, C. elegans amphid sheath (AMsh) glia apical-like domains contact 12 neuron-endings. At these ad-neuronal membranes, AMsh glia localize the K/Cl transporter KCC-3 to a microdomain exclusively around the thermosensory AFD neuron to regulate its properties. Glial KCC-3 is transported to ad-neuronal regions, where distal cilia of non-AFD glia-associated chemosensory neurons constrain it to a microdomain at AFD-contacting glial membranes. Aberrant KCC-3 localization impacts both thermosensory (AFD) and chemosensory (non-AFD) neuron properties. Thus, neurons can interact non-synaptically through a shared glial cell by regulating microdomain localization of its cues. As AMsh and glia across species compartmentalize multiple cues like KCC-3, we posit that this may be a broadly conserved glial mechanism that modulates information processing across multimodal circuits.

Summary:

INTRODUCTION: Across both central and peripheral nervous systems (CNS and PNS, respectively), each glia contacts multiple neurons. This raises a fundamental logic question: does one glial modulate all of its contacting neurons similarly or through distinct mechanisms? Decoding this logic of specificity in glio-neuron interactions molecularly is critical to understand the organizational principle of the nervous system. This is relevant for both CNS and PNP/sensory biology.

RESULTS: The AMsh glia localize the K/Cl transporter KCC-3 to a microdomain around only AFD-NRE. This microdomain is distinct from the glial ad. The glioblastoma-associated neuron contact site is distinct and distinct from other glially associated NREs. The KCC-KCC-3 microdomain terminates at the apical glium. The AFD/NRE contact site and the proximal glius are distinct.

DISCUSSION: Our results provide single-molecule evidence that a single glial cell can differently regulate associated neurons. Study of the underlying biology suggests that neurons can interact with each other non-synaptically by regulating cues on shared glia. Using AMsh glio as a powerful single-glia experimental platform, we show that the glian apical-like ad-neuronal membrane is partitioned into multiple and distinct molecular microdomains around individual NRE contact sites. Further, a pical adenoid membranes extend into a boundary domain we term GAB. This is distinct from classic epithelial aeolian aphasic-basal polarity.

Figures:

Figure 1: KCC-3 localizes specifically around AFD-NRE

Figure 2: KCC-3 localizes to a glial apical microdomain in an age-dependent manner

Figure 3: Microdomains as a general feature of glia

Figure 4: Glial KCC-3 localizes in a two-step process through two protein regions

Figure 5: Glial KCC-3 localization is regulated by distal non-AFD-NRE cilia

Figure 6: Microdomain localization of KCC-3 regulates AFD shape

Figure 7: Schematic of KCC-3 localization in AMsh glia

Mentioned biomedical entities:

Genes: AJM-1, AP, ASE, B, Bai, CEH-36, DAF-6, DYF-11, GCK, GLT-1, KCC-2, KCC-3, LIT-1, Max, ODR-10, SAX-7, TAX-2, TM6, VAP-1, WNK, WNK-1, ceh-37, odr-7, snx-1, ttx-1, unc-101

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, glial cell, neuron

Organisms: Alex, C. elegans, Fisher

Tissues: neuron

Pathways: N/A

Query Result 25/25

Plasticity in gustatory and nociceptive neurons controls decision making in C. elegans salt navigation

Dekkers, Martijn P. J.; Salfelder, Felix; Sanders, Tom; Umuerri, Oluwatoroti; Cohen, Netta; Jansen, Gert

Communications Biology

Year: 2021 PMID: 34504291 Citations: 8 Score: 3.24 DOI: 10.1038/s42003-021-02561-9

Keywords Hits: lsy-6: 2, che-1: 9

Keywords Frequency (TF): lsy-6: 0.000; che-1: 0.001

Abstract:
A conventional understanding of perception assigns sensory organs the role of capturing the environment. Better sensors result in more accurate encoding of stimuli, allowing for cognitive processing downstream. Here we show that plasticity in sensory neurons mediates a behavioral switch in C. elegans between attraction to NaCl in nave animals and avoidance of NaCl in preconditioned animals, called gustatory plasticity. Ca2+ imaging in ASE and ASH NaCl sensing neurons reveals multiple cell-autonomous and distributed circuit adaptation mechanisms. A computational model quantitatively accounts for observed behaviors and reveals roles for sensory neurons in the control and modulation of motor behaviors, decision making and navigational strategy. Sensory adaptation dynamically alters the encoding of the environment. Rather than encoding the stimulus directly, therefore, we propose that these C. elegans sensors dynamically encode a context-dependent value of the stimulus. Our results demonstrate how adaptive sensory computation can directly control an animals behavioral state.

Summary:
Introduction: Adaptive behavior refers to the ability of animals to change their actions in response to changes in the environment or in their internal state. Here, we study a form of short-term sensory adaptation, called gustatory plasticity. Its defining feature in the nematodeCaenorhabditis elegansis the dynamic balance of salt (NaCl) attraction and avoidance as a function of experience. Often, the goal of behavior is implicit (e.g., tracking an animal and recording its neural activity may not disclose what form of reward the animal seeks as it acts in its environment). Furthermore, the advantage gained by Adaptative behavior refer to the able of animals a change their behaviors in response t o changes in th e environment or t

Results: The Ca2reporter Yellow Cameleon identifies the range of nave sensory responses to NaCl. These responses are a result of a gradual increase in the fraction of animals that respond to a 3s exposure to different concentrations of NaCr. The Ca2-response index (RI) is a measure of the proportion of animals responding to osmotic stimuli. The response index is derived from the fraction that responded to varying concentrations.

Discussion: In nave C. elegans, attractive, ASE mediated, and aversive, ASH mediated salt responses are controlled by clearly delineated subcircuits, resulting in a switch between attraction up to 200mM NaCl and avoidance of higher concentrations (Fig). However, while nave animals are attracted to low salt concentrations, extended NaCr exposure without food leads animals to avoid any NaCc concentration implying an adaptive foraging behavior. Here, we showed that the behavioral switch between NaCla attraction and avoidANCE is mediated by plasticity in sensory neurons, causing altered dynamic ranges in both attractive and nociceptive subcircircuit. Based on our experimental and modeling results we propose that the sensory response ofC. eliota to NaCle is regulated at multiple levels.

Reporting summary: The following information is available in thelinked to this article: . Further information on research design is availablein thelinkedto this article. . Supplementary information: - . -. .- .-.-.. / --.-- , : () /-, -,,-(-)-/,

Figures:
Figure 1: Ca 2+ responses of ASEL, ASER, and ASH neurons to brief NaCl exposure.

Figure 2: Prolonged exposure to NaCl sensitizes ASER.

Figure 3: Pre-exposure to NaCl desensitizes ASEL.

Figure 4: Prolonged exposure to NaCl sensitizes ASH.

Figure 5: ASER sensitization is affected by unc-13 and eat-4 mutations.

Figure 6: ASH sensitization requires ASE, glutamate, neuropeptides, dopamine, and serotonin.

Figure 7: Computational model reproduces sensory responses to NaCl in virtual quadrant assay.

Figure 8: Stochastic recruitment of ASH drives gustatory plasticity in our computational model.

Figure 9: Sensory adaptation controls exploration in a virtual spot assay.

Figure 10: Schematic model of the NaCl navigation circuit.

Mentioned biomedical entities:
Genes: ASE, ASER, C, RI, eat-4, flp-6, gpc-1, lsy-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: cell, interneuron, neuron

Organisms: N/A

Tissues: capillary, neuron

Pathways: N/A